

EnergyGaugeSummit Help

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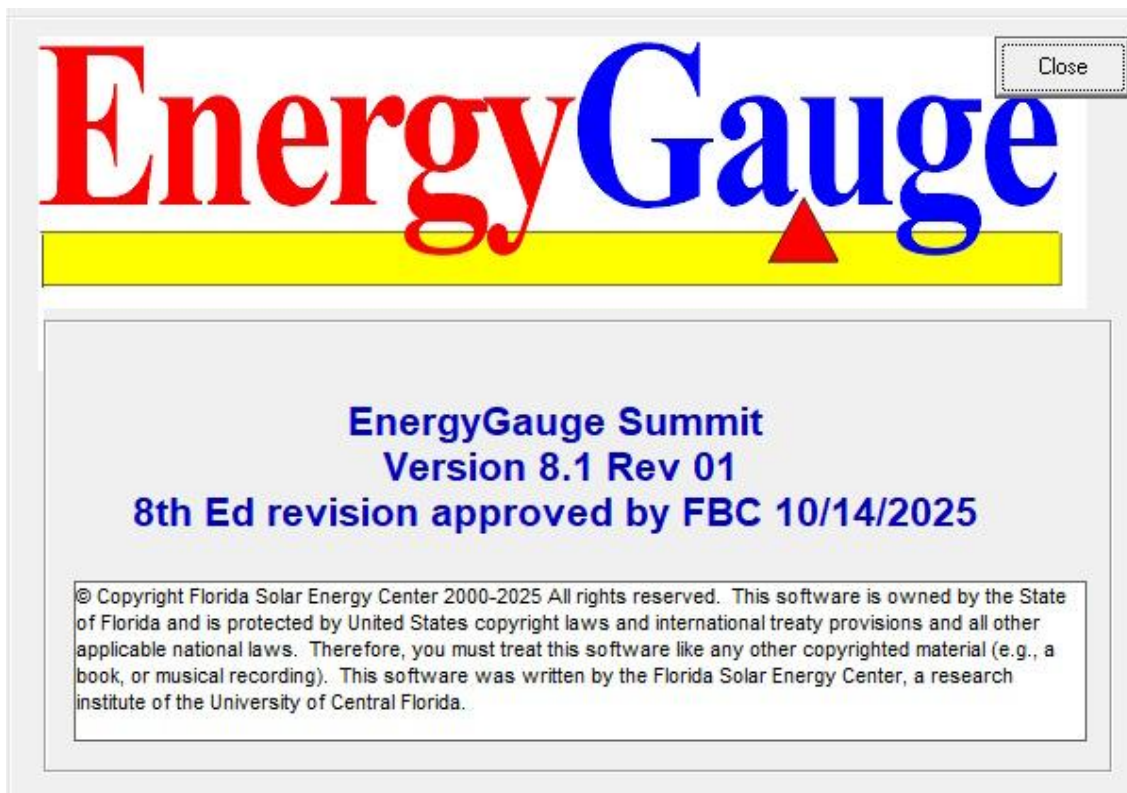
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1. Welcome

1.1 EnergyGauge Summit



EnergyGauge Summit is the Energy Code compliance tool for commercial buildings in Florida and other states in the USA. It was developed by the Florida Solar Energy Center. EnergyGauge Summit is a Windows-based product and will operate under most Microsoft Windows® versions.

The Florida Code version of the software has the following Florida Building Code, Energy Conservation compliance versions:

- 2023 Florida Building Code, Energy Conservation (8th Edition)
- 2020 Florida Building Code, Energy Conservation (7th Edition) and its 2022 supplement
- 2017 Florida Building Code, Energy Conservation (6th Edition)
- 2014 Florida Building Code, Energy Conservation (5th Edition)
- 2010 Florida Building Code, Energy Conservation

- 2008 Florida Building Code, Energy Conservation

In addition, several versions of ASHRAE and IECC national code compliance, ASHRAE Appendix G rating, LEED, and Commercial Building Tax-credit calculation are also available in the Premier version of the software

If you are a new user, it is recommended that you start with the introduction.

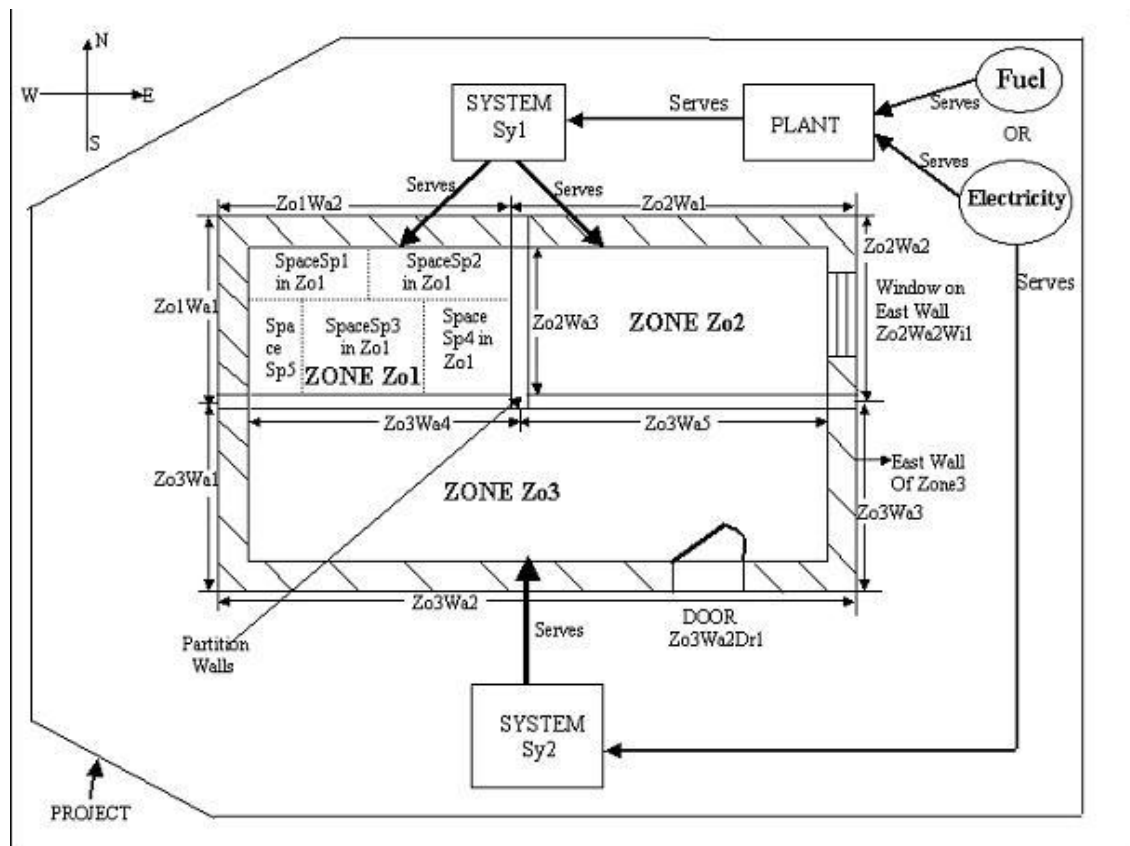
For the latest information, visit

www.energygauge.com

1.2 Introduction

Project Hierarchy

A fundamental concept in EnergyGauge Summit is that it structures the building into a hierarchical collection of elements. For example, a building (sometimes referred to as Project) consists of multiple Zones which in turn may contain many Spaces, Walls, Floors, Roofs etc. Walls can further contain multiple Windows and Doors. The figure demonstrates the concept.



In the illustration, the Project is composed of the building, systems and the plant.

The project has a PLANT, that may include chillers and/or boilers that are served by the utility and/or other fuel sources.

The project has two systems, Sy1 and Sy2

System Sy1 is served by the PLANT

System Sy2 is a packaged system type that is served directly by the utility.

The building consists of three zones: Zo1; Zo2; and Zo3. (A Zone is a unit of a building in which the air is assumed to be well mixed and where the temperature can be assumed to be uniform)

System Sy1 serves the following Zones: Zo1 and Zo2

System Sy2 serves only Zone Zo3

Zone Zo1 consists of five spaces of various sizes: Sp1; Sp2; Sp3; Sp4; and Sp5. The sum of the areas and volumes of the five spaces constitute the area and volume of Zo1. (A space is a portion of a Zone having a distinct internal load characteristic, such as Lighting and Equipment loads)

Zone Zo1 has two exterior walls: Zo1Wa1 (west) and Zo1Wa2 (north).

Zone Zo2 has two exterior walls: Zo2Wa1 (north) and Zo2Wa2 (East), and a partition wall Zo2Wa3 that is next to Zone Zo1.

The east wall Zo2Wa2 in zone Zo2 contains a window Zo2Wa2Wi1. The orientation of windows need not be specified. It is inherited from the wall on which it is mounted.

Zone Zo3 has three exterior walls namely Zo3Wa1 (west), Zo3Wa2 (south), and Zo3Wa3 (east).

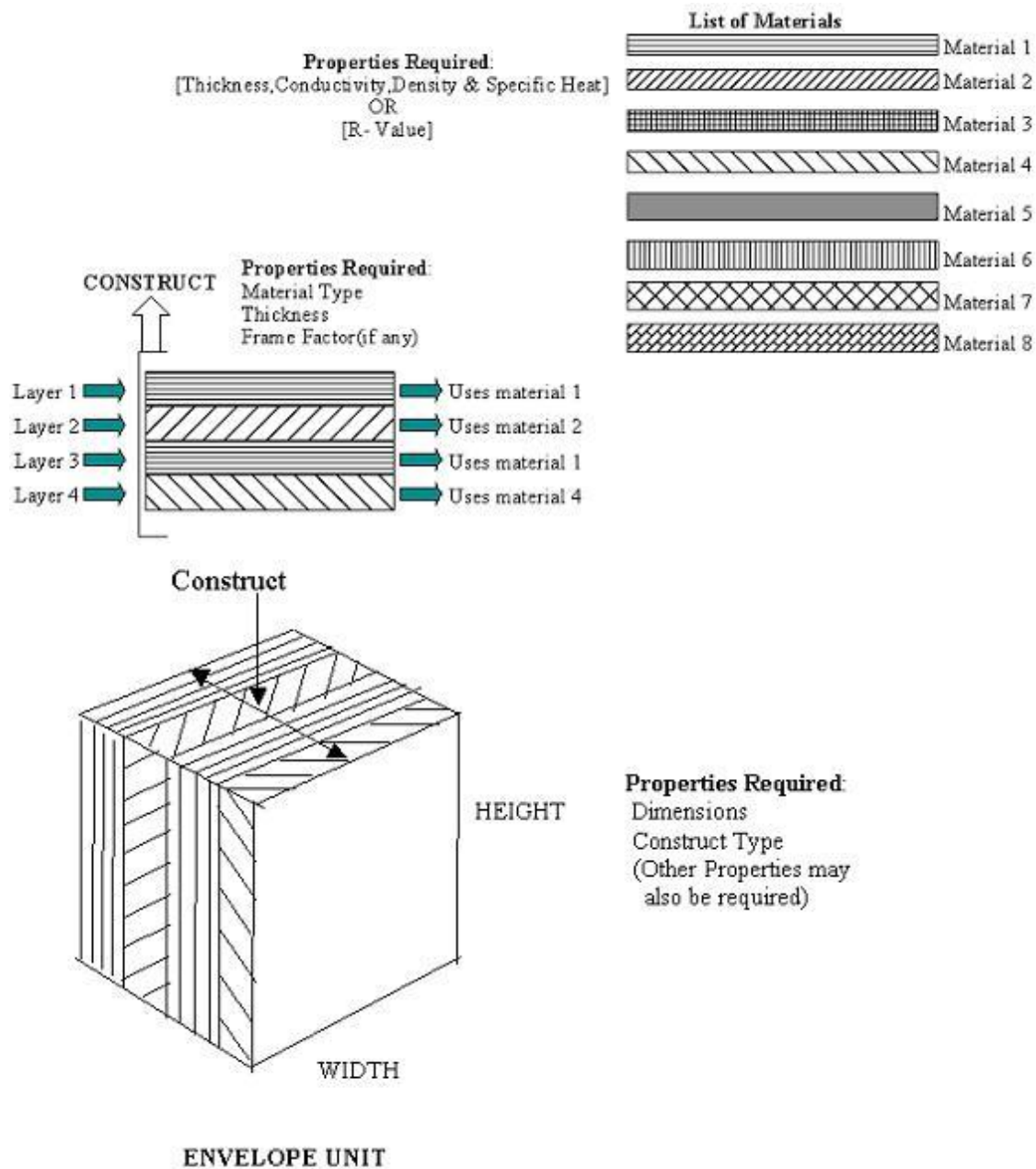
Zone Zo3 has two partition walls: Zo3Wa4 which is next to Zone Zo1 and another Zo3Wa5 that is next to zone Zo2.

The south wall Zo3Wa2 in zone Zo3 contains a door Zo3Wa2Dr1. The orientation of door need not be specified. It is inherited from the wall on which it is mounted.

Materials and Constructs - Introductory Concepts

Unlike prior DOS versions of Summit where choices for construct (or envelope types) were mostly limited to prebuilt Wall or Roof types, EnergyGauge Summit gives the user virtually unlimited choices. First, the software provides the user with an extensive list of constructs for walls, roofs and floors. Second, the user has the option of creating

their own constructs and using them repeatedly. Therefore an understanding of the basics behind constructs is crucial in being able to use the software effectively.



A building may contain several envelope units such as walls, roofs and floors.

Each envelope unit has associated dimensions (such as width and height) and a construct.

The construct specifies the physical make up of the envelope unit.

More than one envelope unit can have the same construct. Therefore, once a construct is developed or included in a project, it can be used to specify the make up of several envelope units.

A construct consists of one or more layers that make up the construct, except for a simple construct where layers do not exist and gross properties of the construct are used in the calculations

Each layer is a slice of the construct that has distinct characteristics that include the material it is made of, thickness and framing factor (if any).

The aggregate of each property of the layers constitutes the overall property of the construct. For example, the sum of the thickness of the layers is the total thickness of the construct.

Each layer must be associated with a material that must exist prior to building a construct.

A material has specific detailed thermal properties such as conductivity, density, specific heat and thickness. Alternatively, in lieu of these properties one can specify only the R-Value in which case the material is assumed to be massless.

For details on how to build constructs, see the section on Project Materials and Constructs

Project

What is a project?

A project is your building input containing components such as zones, walls etc. When saved, the file has an extension .egc

Create a new project?

You can open a project using the **'New'** tool button or from the **'File' > 'New'** menu.

Open an existing project?

You can open a project using the **'Open'** tool button or from the **'File' > 'Open'** menu.

Templates

What is a template?

A template is a file with preset information that can be used as a starting point for a new project. Templates contain preset project and resource information such as constructs, materials and fenestrations. Users may create and save as many templates as they need, and choose any template to create a new Project. Creating new projects from templates can save considerable time, especially if constructs,

materials and fenestrations that are often used are saved in a template for repeated use. In addition, the user can also set the default fenestrations envelope constructs.

Create a template?

From the main menu, Click '**File**' > '**New**' > '**Template**'. Make your modifications and save it.

Modify an existing template?

Click 'Open'. Then select a template (a file with .EGT extension) from the open dialog. Make your modifications and save it.

Create a new project using templates?

1. A template file must first exist. If not, create one. Or use the sample template provided with the software.
2. Click '**File**' > '**New**' > '**Project from Template**'.
3. Select a template (a file with .EGT extension) from the open dialog.
4. This creates a new project based on the selected template.

How do I know if I have opened a template or a project file?

A Project will have a Project Explorer (tree). A template does not have an explorer (tree). In addition, the form layout will be significantly different for each. See the figure below for an example of a template layout.

To modify Fenestrations in the template click '**Tmpl Fenestrations**' button located on the right. Make your modifications. An alternate access is through the main menu. Click '**Edit**' > '**Template Fenestrations**'.

To modify Materials and Constructs in the template click '**Tmpl Materials and Constructs**' button located on the right. Make your modifications. See the section on Materials and Constructs Library for editing functions. An alternate access is through the main menu. Click '**Edit**' > '**Template Materials and Constructs**'.

To set or modify preferences, in the main menu click '**Edit**' > '**Template Preferences**'. and make your modifications.

Master Library

What is the Master Library?

The Master Library contains extensive resources on materials, fenestration and constructs. It serves as a repository from which you can draw into your individual project or template. Users cannot change information contained in the master library.

Accessing the Master Library

From the main menu, click '**Resources**' --> '**Master Library**'. Then click the library component you want to open.

Note: The Master Library can be accessed even if a project is not open.

Project Library

What is a Project Library?

The project library is similar to the master library but unlike the Master Library is associated with a specific project and provides resources for the project. Each project has its own project library. Project Libraries may be edited.

Accessing the Project Library

First, a project must be open. You can access the Project Library by selecting the "**Prj Resources**" tab of the project explorer and then clicking the library button you want access. Alternatively, you can access it from the main menu. To do this from the main menu click '**Edit**'. Then click the project library component you want to open.

Note: The project library can be accessed only if a project is open.

Miscellaneous

Some fields may be inactive, why?

Some fields are intentionally kept inactive or not editable. They usually represent the defaults used in calculations and are not user-editable.

What are tool tips? How do I get tool tips displayed?

In general, if you hover the mouse on any (non grid) field on any form, a tool tip appears giving more information about that field, such as data ranges and units.

What are the units of the various data?

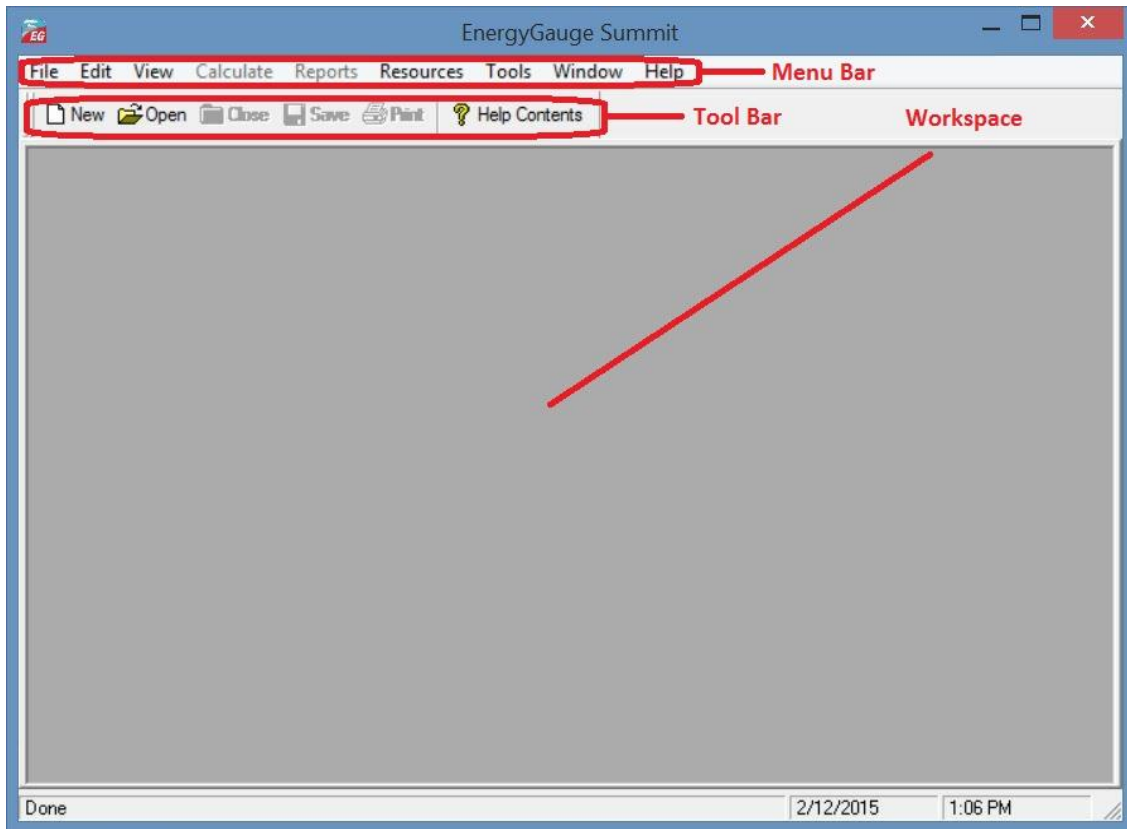
Units are not written on the forms. Use the tool tip feature above to check input units.

Important note:

You must tab out of the field you are editing (or move to another field and click the mouse) before the change will take effect. If you switch screens or choose to calculate or save without first moving to another field, the change will not be saved or included in the calculation.

General Layout

Upon starting EnergyGauge Summit, the following window comes up.



To create a new project click the '**New**' tool button.

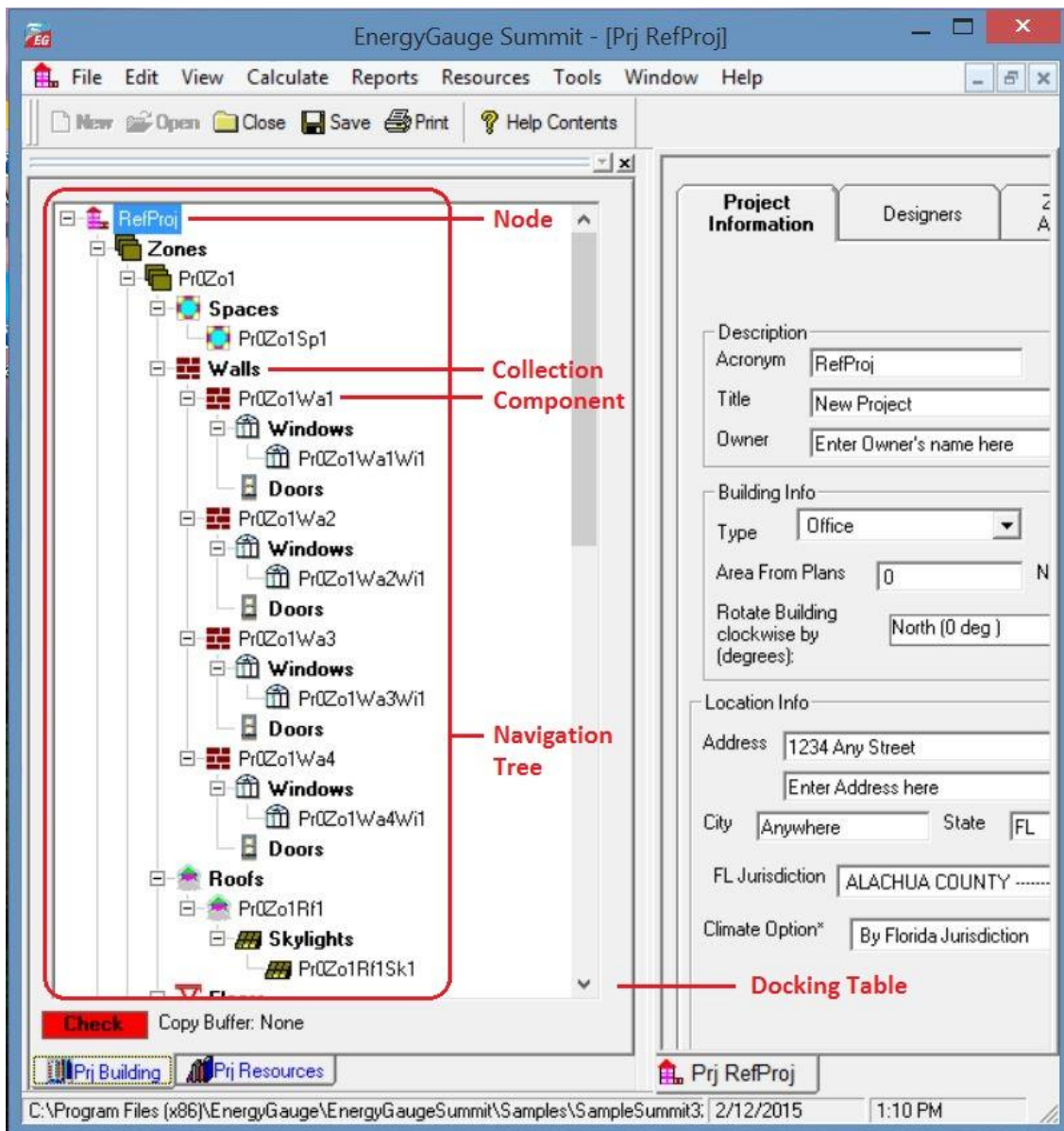
To open and work on an existing project, click the '**Open**' tool button.

You can also open an existing or new project from the '**File**' menu.

If you require help, click on the '**Help Contents**' button or use the '**Help**' from the main menu.



1.3 Project Explorer



When a project is opened, the Project Explorer (sometimes called building tree) appears. It is a hierarchical representation of the entire building. It is through the Project Explorer that one navigates the various building components. It is also used to add and delete various building components.

In general, the Project Explorer (tree) consists of nodes representing Collections and Components. Collections are containers that appear in bold face and contain one or more components under them. Example: the 'Zones' Collection contains one or more individual zones. A component represents a physical element of a building such as a wall, door, or window etc.

Collections

A collection is a group of a particular type of component. All the components in a collection are of the same type. For example, a collection of Zones only consists of individual zones.

Individual Components

A component represents a single building element. One or more components of the same type together form a collection.

Adding a component

To add any new component of a particular type first highlight the collection by left-clicking it. Right-click on the selected collection and use the 'add' option from the pops-up menu. A new component of that type will be added and the input data form will appear (usually) on the right side of the screen.

Example: to add a new wall under Zone 1, first select 'Walls' under Zone 1. Keep the cursor on the 'Walls' and right click. Select the 'Add New' option from the pop-up menu to add a new wall. The data input form for the wall will show up on the right. You may now input your data for the new wall.

Selecting a component to edit

Just clicking on an individual component opens the data input form for that component, For example, clicking on the component labeled 'Zone 1' opens the data input form for Zone 1.

Deleting a component

Select (left-click) an individual component node. Right-click the component and choose 'Delete' from the pop-up menu.

Expanding and collapsing the Project Explorer tree

Click on the + or - sign on the tree node to expand or collapse the tree. Right-clicking on a selected node opens a pop-up menu with options to expand or collapse the tree.

Exceptions - 'Plant', 'Water Heating', 'External Lighting' and 'Piping' are Collections whose components are placed on a form grid in a table format. As an example, click 'Water heating'. An input data form containing a grid with all water heaters in the project will appear.

Copying and Pasting

You can copy any component of a building, such as a Zone or Wall and paste it into the corresponding collection. For example a 'Wall' can be copied and then pasted into any 'Walls' collection.

Steps to Copy and Paste:

- 1) Select (using the mouse) the component that is to be copied
- 2) Right-click on the selected component. A pop-up menu appears.
- 3) Select the copy option in the pop-up menu. (The component is now placed in a Copy buffer indicated by the label at the bottom of the Project Explorer)
- 4) Select (using the mouse) the corresponding Collection where you would like to paste the copied component. For example, if you copied a Wall, select any 'Walls' collection.
- 5) Right-click on the selected collection. A pop-up menu appears.
- 6) Select the 'Paste' option in the pop-up menu. The component will be copied into the selected collection.



1.4 Working with Grids

Working with grids (or Data Tables) - General Procedures

Grids (or data tables) are extensively used in EnergyGauge Summit. These spreadsheet like grids can display large amounts of data in a concise fashion in addition to providing the ability to edit displayed data. They appear on several forms, such as Project and Master Libraries, '*Water Heating*', '*External Lighting*', '*Piping*', '*Space*', etc. A clear understanding of how to work with grids is critical in effectively using the software. The general actions that the user is likely to perform with grid are described in this section.

Constructs				Materials			
	Mat No	Desc	Conductivity [Btu/h.ft.F]	Density [lb/cf]	SpecificHeat [Btu/lb.F]	Thick [ft]	R-Value [h.s.f.F/Btu]
1	264	ALUMINUM, 1/16 IN	26.0000	480.0000	0.1000	0.0050	0.0002
2	214	POLYSTYRENE, EXP., 1-1/4IN	0.0200	1.8000	0.2900	0.1042	5.2100
3	187	GYP OR PLAS BOARD, 1/2IN	0.0920	50.0000	0.2000	0.0417	0.4533
4	178	CARPET W/RUBBER PAD	0.0000	0.0000	0.0000	0.0000	1.2300
5	265	Soil, 1 ft	0.5000	100.0000	0.2000	1.0000	2.0000
6	48	6 in. Heavyweight concrete	1.0000	140.0000	0.2000	0.5000	0.5000
7	267	0.75" stucco	0.4000	16.0000	0.2000	0.0625	0.1563
8	215	POLYSTYRENE, EXP., 2IN	0.0200	1.8000	0.2900	0.1667	8.3350
9	105	CONC BLK HW, 8IN, HOLLOW	0.6060	69.0000	0.2000	0.6667	1.1002
10	256	WOOD, SOFT, 1-1/2IN	0.0660	32.0000	0.3300	0.1250	1.8939
11	86	BRICK, COMMON, 4IN	0.4160	120.0000	0.2000	0.3333	0.8012
12	23	6 in. Insulation	0.0250	5.7000	0.2000	0.5000	20.0000
13	4	Steel siding	26.0000	480.0000	0.1000	0.0050	0.0002
14	271	2x4@24" oc + R11 Batt	0.0280	7.1100	0.2000	0.2917	10.4179
15	94	BUILT-UP ROOFING, 3/8IN	0.0930	70.0000	0.3500	0.0313	0.3366
16	21	4 in. Insulation	0.0250	5.7000	0.2000	0.3330	13.3200
17	9	Air space resistance	0.0000	0.0000	0.0000	0.0000	0.9100

Typical Data Grid Functions

Add rows to grids

There are generally two ways to add rows to grids.

- 1) Select the grid (just click anywhere on the grid) using the mouse and then press the **'Insert'** key on the keyboard. Follow the instructions that appear.
- 2) Right-click the mouse over the grid and then select the **'Add'** option from the pop-up menu. Follow the instructions that appear.

Delete a rows on a grid

There are generally two ways to delete rows from a grids.

- 1) Select any cell on the row you want to delete using the mouse and then press the **'Delete'** key on the keyboard. Follow the instructions that appear.
- 2) Right-click on the cell in the row you want to delete on the grid and then select **'Delete'** from the Pop up menu. Follow the instructions that appear.

Move between cells on a grid

Use the mouse to click on a cell. Also, you can use the Tab key to move from cell to cell.

Edit individual grid cells

There are several types of grid cells.

Cells with numeric or text data

These are cells where numeric or text data can be entered. Double-clicking the cell puts it in edit mode. Pressing the F2 key also has the same effect. You can then enter the new value. To complete the edit, press the **'Enter'** key on the keyboard or click elsewhere on the grid. If the cell is already active (as shown by a light focus rectangle), just begin typing the new value to edit the cell.

Cells where appears when selected

These cells usually require data from other dialogs. Click on the  button and follow the instructions that appear.

Cells where appears when selected

These cells require selection from the drop down. Click on the  button and make your selection.

Cells where appears when selected

Values in these cells can be changed up or down. Click on the up or down arrow depending on what you want to do.

Cells where ☐ or ☒ appear

These specify an unselected or selected option. Values in these cells can be toggled by clicking on the cell, or pressing your **'Space-Bar'** key on your key board.

Note: Some grid cells may not allow any of the above. It is usually because the cells are temporarily or permanently non-editable.

Sorting grids

Most grids can be sorted by clicking the column header. To sort a single column, click on that column header. Clicking a header on an already sorted column reverses the sort order and resorts. You can sort multiple columns (up to three) by holding down the **'CTRL'** key and clicking up to three column headers one after another (while keeping then **'CTRL'** key down).

Manually or automatically (Auto size) resize grid column widths

In general, double clicking on a column header will auto size the column width to the largest entry in that column. To auto size all columns, double click the top left corner label that is common to both the column headers and row labels. You can manually resize column widths by moving your mouse between column headers until a resize cursor appears. Hold the left mouse down and move left or right.

Manually resize Grid Row heights

You can manually resize row height by moving the mouse between row labels until a resize cursor appears. Hold the left mouse down and move up or down.



1.5 Steps to Comply with Florida 8th Ed (2023) Building Code

Compliance options

The four compliance procedures for the FLORIDA ENERGY CODE COMPLIANCE FOR COMMERCIAL BUILDINGS are:

1. ASHRAE 90.1-2019 - Energy Cost Budget Method
2. ASHRAE 90.1-2019 - Envelope Trade-Off Option (**Not supported**)
3. ASHRAE 90.1-2019 - Prescriptive Compliance Option
4. ASHRAE 90.1-2019 - Appendix G Performance Rating Option
5. FBC 2023 - Total Building Performance Compliance Option
6. FBC 2023 - Prescriptive Compliance Option
7. FBC 2023 – Component Performance Alternative Option

These can be invoked by going to the appropriate items in the calculate menu as follows:

'Calculate' -> '8th Edition (2020) Florida Energy Code Compliance' -> 'Total Building Performance Compliance Option' or other options

Steps for compliance:

1. Open new file and enter data. The main user manual explains all the screens, input procedures and input details.
2. After entering inputs go to Calculate -> Error Check. Correct errors that are found
3. Run the appropriate compliance Calculate -> 8th Edition (2023) Florida Energy Code Compliance -> Total Building Performance Compliance Option or other options
4. Check the reports generated to see if the building complies. See section on Appendix A on input and output reports.
5. Print the reports and look for elements that have failed. The reasons for failing will be given in the reports
6. Make changes to inputs that have failed and re run the calculations until you are able to comply.

New Topic (*Press Shift-Enter to insert a line break. Press Enter to add a paragraph return.*)

1.6 Steps to Comply with Florida 7th ed (2020) Building Code

Compliance options

The four compliance procedures for the FLORIDA ENERGY CODE COMPLIANCE FOR COMMERCIAL BUILDINGS are:

1. ASHRAE 90.1-2016 - Energy Cost Budget Method
2. ASHRAE 90.1-2016 - Envelope Trade-Off Option (Not supported)
3. ASHRAE 90.1-2016 - Prescriptive Compliance Option
4. ASHRAE 90.1-2016 - Appendix G Performance Rating Option (Not supported)
5. IECC 2018 - Total Building Performance Compliance Option
6. IECC 2018 - Prescriptive Compliance Option
7. IECC 2018 – Component Performance Alternative Option

These can be invoked by going to the appropriate items in the calculate menu as follows:

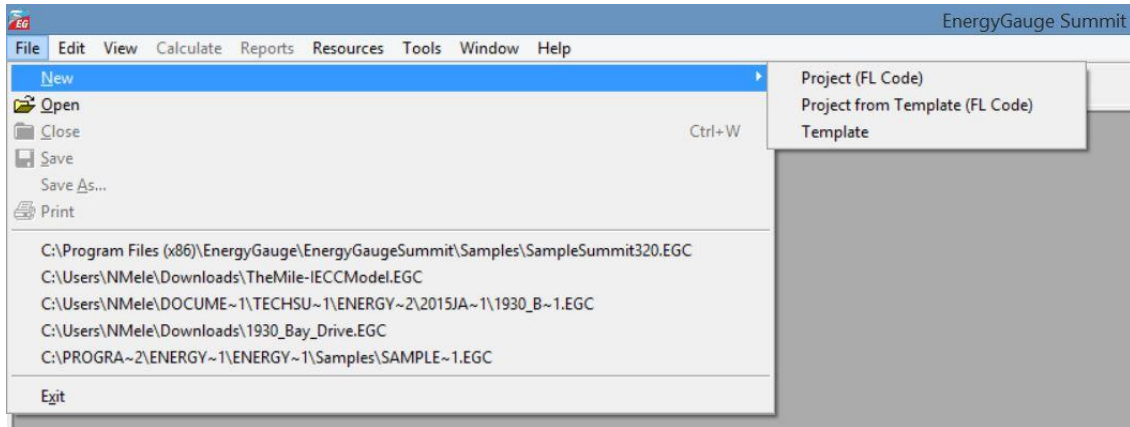
'Calculate' -> '7th Edition (2020) Florida Energy Code Compliance' -> 'IECC 2018 - Total Building Performance Compliance Option' or other options

Steps for compliance:

1. Open new file and enter data. The main user manual explains all the screens, input procedures and input details.
2. After entering inputs go to Calculate -> Error Check. Correct errors that are found
3. Run the appropriate compliance Calculate -> 7th Edition (2020) Florida Energy Code Compliance -> IECC 2018 - Total Building Performance Compliance Option or other options
4. Check the reports generated to see if the building complies. See section on Appendix A on input and output reports.
5. Print the reports and look for elements that have failed. The reasons for failing will be given in the reports
6. Make changes to inputs that have failed and re run the calculations until you are able to comply.

2. Top Menus

2.1 File Menu



The **File** menu consists of the following sub menus:

New - It consists of 3 sub menus.

Project(FL Code): Opens a new project for code compliance.

Project From Template (FL Code): Opens a new project for code compliance from a template. When this option is chosen, a dialog appears for selecting the template the new project is to be based on.

Template: Creates a new Template.

Open - Opens a file. Either a project (.EGC) file or a template (.EGT) file can be opened for data input.

Close - Closes the current file.

Save - Saves the changes made to the current file on to the hard disk.

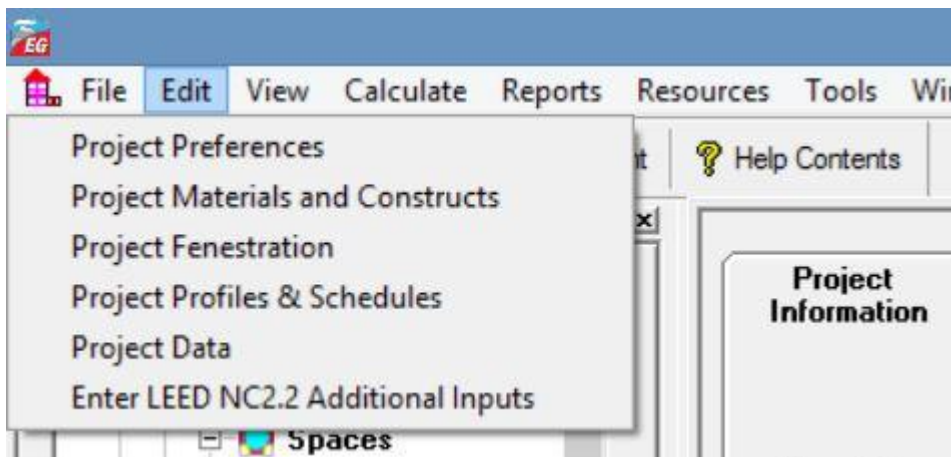
Save as - Saves the current file under a new user specified filename.

Print - Prints the input data of a currently open project.

Exit - Closes the application.

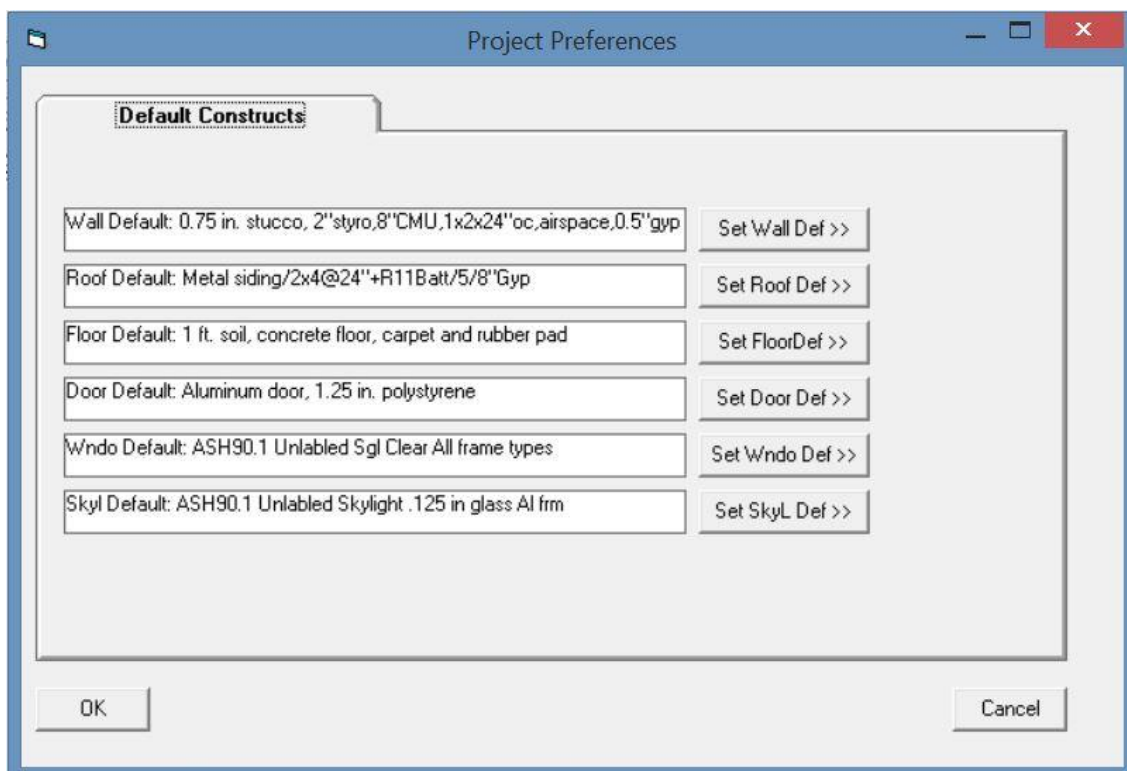


2.2 Edit Menu



The 'Edit' menu has the following sub menus:

Project Preferences - This sub menu opens the dialog for settings defaults for the current project or template as shown below:



You can set the default type for each envelope component. Any new envelope component you add to the project will use these settings. You can change these setting at any time.

Project Materials and Constructs - This sub menu opens the Project Materials and Constructs Library for editing.

Project Fenestration - This sub menu open the Project Fenestration Library for editing.

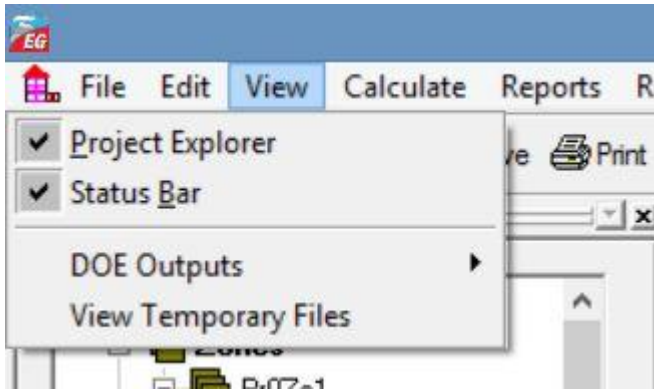
Project Data - This sub menu opens the [Project Data Input Form](#) for editing.

Enter LEED 2.2NC Additional Inputs - This sub menu opens the [LEED NC22 Additional Inputs Form](#) for editing.

Project Profiles & Schedules - This sub menu opens the [Project Profiles & Schedules](#) menu for editing.



2.3 View Menu



The 'View' menu contains the following sub menus:

Project Explorer - Shows or hides the Project Explorer as applicable.

Status Bar - Displays an informational status bar at the bottom when checked.

DOE Outputs - Has the following sub menus:

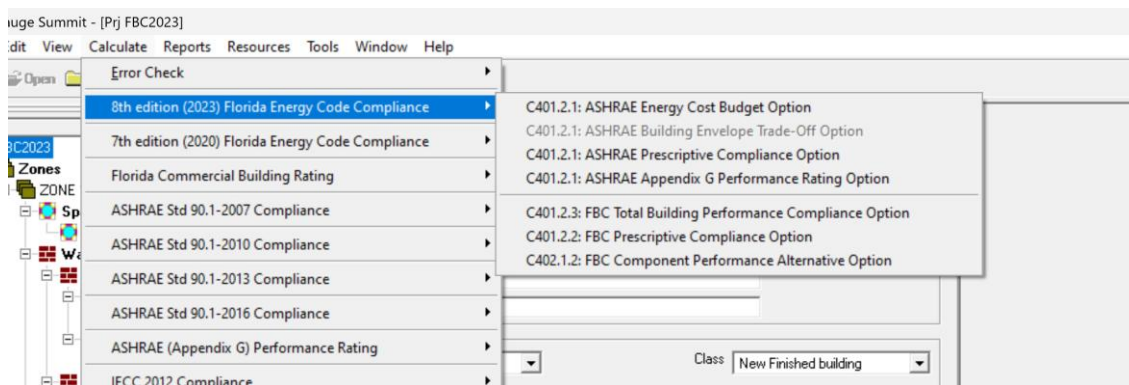
- * *Last Design Run*

- * *Last Run*

The calculation engine, DOE-2.1E, produces text outputs that can be viewed for possible errors, warnings and cautions. The above two sub menus are provided only to be used for debugging should the need arise.



2.4 Calculate Menu



The '**Calculate**' menu has the following sub menus:

Error Check - Displays the data errors in the project. It usually consists of the component being error-checked, an error message, possible cause, and possible resolution. It is recommended that an error check be performed before a compliance calculation is run. **You can also check for errors by clicking the red 'Check' label at the bottom of the Project Explorer. Nodes with errors appear in red. Hovering the mouse on the node with error displays a tool tip with the error message.**

- * *Whole Building* - Checks errors for the whole building.
- * *Envelope Only* - Checks errors only for the building envelope namely the walls, floors and roof.

7th Edition (2020) Florida Energy Code Compliance - Compliance reports will be displayed on the screen when **Calculate -> Compliance** and one of the following options is chosen. The Compliance sub menu has the following options:

- * *C402.1.1: ASHRAE Energy Cost Budget Option* - Runs the Energy Cost Budget Method described in ASHRAE 90.1-2016. This is a computer based annual energy performance calculation. Under this method, energy performance is calculated for the entire building based on the envelope and major energy-consuming systems specified in the design, and simultaneously for a reference building of the same configuration but with reference features.
- * *C402.1.1: ASHRAE Building Envelope Trade-Off Option* *(Not supported at this time)* .
- * *C402.1.1: ASHRAE Prescriptive Compliance Option* - Runs the Prescriptive Compliance Option described in ASHRAE 90.1-2016. Under this method, the building envelope must meet minimum prescriptive criteria.
- * *C402.1.1: ASHRAE Appendix G Performance Rating Option* - Runs the Appendix G performance Rating Method described in ASHRAE 90.1-2016 - App G. This is a

computer based annual energy performance calculation. Under this method, energy performance is calculated for the entire building based on the envelope and major energy-consuming systems specified in the design, and simultaneously for a baseline building of the same configuration but with baseline features. *(Not supported at this time)*

* *C402.1.3: FBC Total Building Performance Compliance Option* - Runs the Total Building Performance Compliance. This is a computer based annual energy performance calculation. Under this method, energy performance is calculated for the entire building based on the envelope and major energy-consuming systems specified in the design, and simultaneously for a reference building of the same configuration but with reference features.

* *C402.1.2: FBC Prescriptive Compliance Option* - Runs the Prescriptive Compliance Option Under this method, the building envelope must meet minimum prescriptive criteria.

* *C402.1.5: FBC Component Performance Alternative Option* - Runs the Component Performance Alternative Option. Under this method, trade-off is allowed between envelope components.

8th Edition (2023) Florida Energy Code Compliance - Compliance reports will be displayed on the screen when **Calculate** -> **Compliance** and one of the following options is chosen. The Compliance sub menu has the following options:

* *C401.2.1: ASHRAE Energy Cost Budget Option* - Runs the Energy Cost Budget Method described in ASHRAE 90.1-2016. This is a computer based annual energy performance calculation. Under this method, energy performance is calculated for the entire building based on the envelope and major energy-consuming systems specified in the design, and simultaneously for a reference building of the same configuration but with reference features.

* *C401.2.1: ASHRAE Building Envelope Trade-Off Option* *(Not supported at this time)* .

* *C401.2.1: ASHRAE Prescriptive Compliance Option* - Runs the Prescriptive Compliance Option described in ASHRAE 90.1-2016. Under this method, the building envelope must meet minimum prescriptive criteria.

* *C401.2.1: ASHRAE Appendix G Performance Rating Option* - Runs the Appendix G performance Rating Method described in ASHRAE 90.1-2019 - App G. This is a computer based annual energy performance calculation. Under this method, energy performance is calculated for the entire building based on the envelope and major energy-consuming systems specified in the design, and simultaneously for a baseline building of the same configuration but with baseline features.

* *C402.2.2: FBC Total Building Performance Compliance Option* - Runs the Total Building Performance Compliance Option described in IECC 2018. This is a computer based annual energy performance calculation. Under this method, energy performance is calculated for the entire building based on the envelope and major energy-consuming systems specified in the design, and simultaneously for a reference building of the same configuration but with reference features.

* *C402.2.3: FBC Prescriptive Compliance Option* - Runs the Prescriptive Compliance Option described in IECC 2018. Under this method, the building envelope must meet minimum prescriptive criteria.

* *C402.2.3: FBC Component Performance Alternative Option* - Runs the Component Performance Alternative Option described in IECC 2018. Under this method, trade-off is allowed between envelope components.

Florida Commercial Building Rating

* *2008 FL Rating* - Runs a simulation that compares the total energy use of the design building under consideration, with the reference building created using the ASHRAE 90.1 (2007, Appendix G) standard. The rating report gives an index called the EPI (Energy Performance Index) that indicates how much the building under consideration costs to run as compared to the baseline reference building.

2007 ASHRAE 90.1 Compliance

* *Energy Cost Budget Method* - Runs the whole building code calculation. This is a computer based annual energy performance calculation. Under this method, energy performance is calculated for the entire building based on the envelope and major energy-consuming systems specified in the design and simultaneously for a reference building of the same configuration, but with reference features. Note that this runs and displays only the Whole building compliance calculation.

* *Envelope Trade-Off Option* - Runs the component code calculation. This is a computer based calculation methodology. Under this method, each component of the building systems must meet minimum performance standards. Note that this runs and displays only the Envelope compliance calculation. *(Not supported at this time)*

* *Prescriptive Building Option* - This option checks for line by line compliance of each element in the building.

2010 ASHRAE 90.1 Compliance

* *Energy Cost Budget Method* - Runs the whole building code calculation. This is a computer based annual energy performance calculation. Under this method, energy performance is calculated for the entire building based on the envelope and major energy-consuming systems specified in the design and simultaneously for a

reference building of the same configuration, but with reference features. Note that this runs and displays only the Whole building compliance calculation.

* *Envelope Trade-Off Option* - Runs the component code calculation. This is a computer based calculation methodology. Under this method, each component of the building systems must meet minimum performance standards. Note that this runs and displays only the Envelope compliance calculation. *(Not supported at this time)*

* *Prescriptive Building Option* - This option checks for line by line compliance of each element in the building.

2013 ASHRAE 90.1 Compliance

* *Energy Cost Budget Method* - Runs the whole building code calculation. This is a computer based annual energy performance calculation. Under this method, energy performance is calculated for the entire building based on the envelope and major energy-consuming systems specified in the design and simultaneously for a reference building of the same configuration, but with reference features. Note that this runs and displays only the Whole building compliance calculation.

* *Envelope Trade-Off Option* - Runs the component code calculation. This is a computer based calculation methodology. Under this method, each component of the building systems must meet minimum performance standards. Note that this runs and displays only the Envelope compliance calculation. *(Not supported at this time)*

* *Prescriptive Building Option* - This option checks for line by line compliance of each element in the building.

2016 ASHRAE 90.1 Compliance

* *Energy Cost Budget Method* - Runs the whole building code calculation. This is a computer based annual energy performance calculation. Under this method, energy performance is calculated for the entire building based on the envelope and major energy-consuming systems specified in the design and simultaneously for a reference building of the same configuration, but with reference features. Note that this runs and displays only the Whole building compliance calculation.

* *Envelope Trade-Off Option* - Runs the component code calculation. This is a computer based calculation methodology. Under this method, each component of the building systems must meet minimum performance standards. Note that this runs and displays only the Envelope compliance calculation. *(Not supported at this time)*

* *Prescriptive Building Option* - This option checks for line by line compliance of each element in the building.

* *Appendix G Performance Rating Option* - Runs the Appendix G performance Rating Method described in ASHRAE 90.1-2016 - App G. This is a computer based annual energy performance calculation. Under this method, energy performance is calculated for the entire building based on the envelope and major energy-consuming systems specified in the design, and simultaneously for a baseline building of the same configuration but with baseline features. *(Not supported at this time)*

ASHRAE (Appendix G) Performance Rating - This specifies a performance rating method and is a modification of the Energy Cost Budget Method specified in ASHRAE Standard 90.1. It is used for buildings that substantially exceed the requirements in Standard 90.1 but is not meant as an alternative to the Energy Cost Budget requirements stated in Standard 90.1. It is useful for evaluating all proposed designs, including alterations and additions to existing buildings, except designs with no mechanical systems.

* *Version 2007* - Runs an Appendix G rating for ASHRAE Standard 90.1 2007

* *Version 2010* - Runs an Appendix G rating for ASHRAE Standard 90.1 2010

* *Version 2013* - Runs an Appendix G rating for ASHRAE Standard 90.1 2013

* *Version 2016* - Runs an Appendix G rating for ASHRAE Standard 90.1 2016 *(Not supported at this time)*

IECC 2012 Compliance

* *Total Building Performance Compliance* - Runs the whole building code calculation. This is a computer based annual energy performance calculation. Under this method, the energy performance is calculated for the entire building and compared to a reference structure generated via the IECC 2012 Code.

* *Prescriptive Compliance* - Runs a component-by-component compliance for each building element based off the IECC 2012 Code.

IECC 2015 Compliance

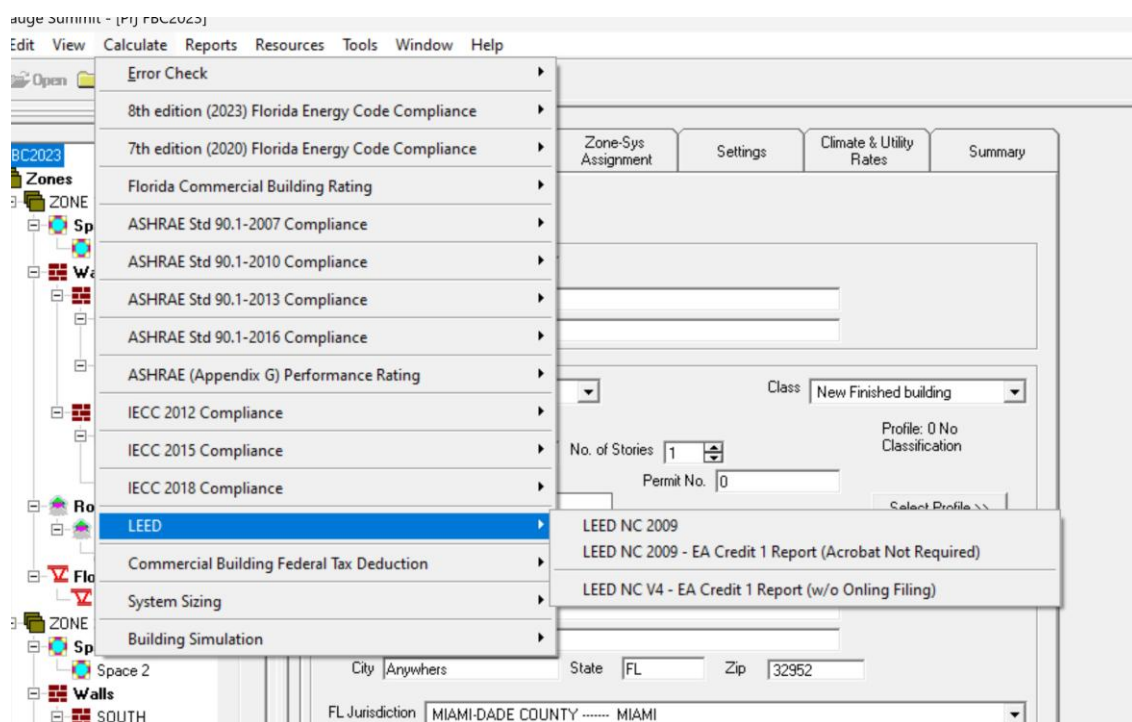
* *Total Building Performance Compliance* - Runs the whole building code calculation. This is a computer based annual energy performance calculation. Under this method, the energy performance is calculated for the entire building and compared to a reference structure generated via the IECC 2015 Code.

* *Prescriptive Compliance* - Runs a component-by-component compliance for each building element based off the IECC 2015 Code.

- * *Component performance Alternative* - Runs the Component Performance Alternative Option described in IECC 2015. Under this method, trade-off is allowed between envelope components.

IECC 2018 Compliance

- * *Total Building Performance Compliance* - Runs the whole building code calculation. This is a computer based annual energy performance calculation. Under this method, the energy performance is calculated for the entire building and compared to a reference structure generated via the IECC 2018 Code.
- * *Prescriptive Compliance* - Runs a component-by-component compliance for each building element based off the IECC 2018 Code.
- * *Component performance Alternative* - Runs the Component Performance Alternative Option described in IECC 2018. Under this method, trade-off is allowed between envelope components.



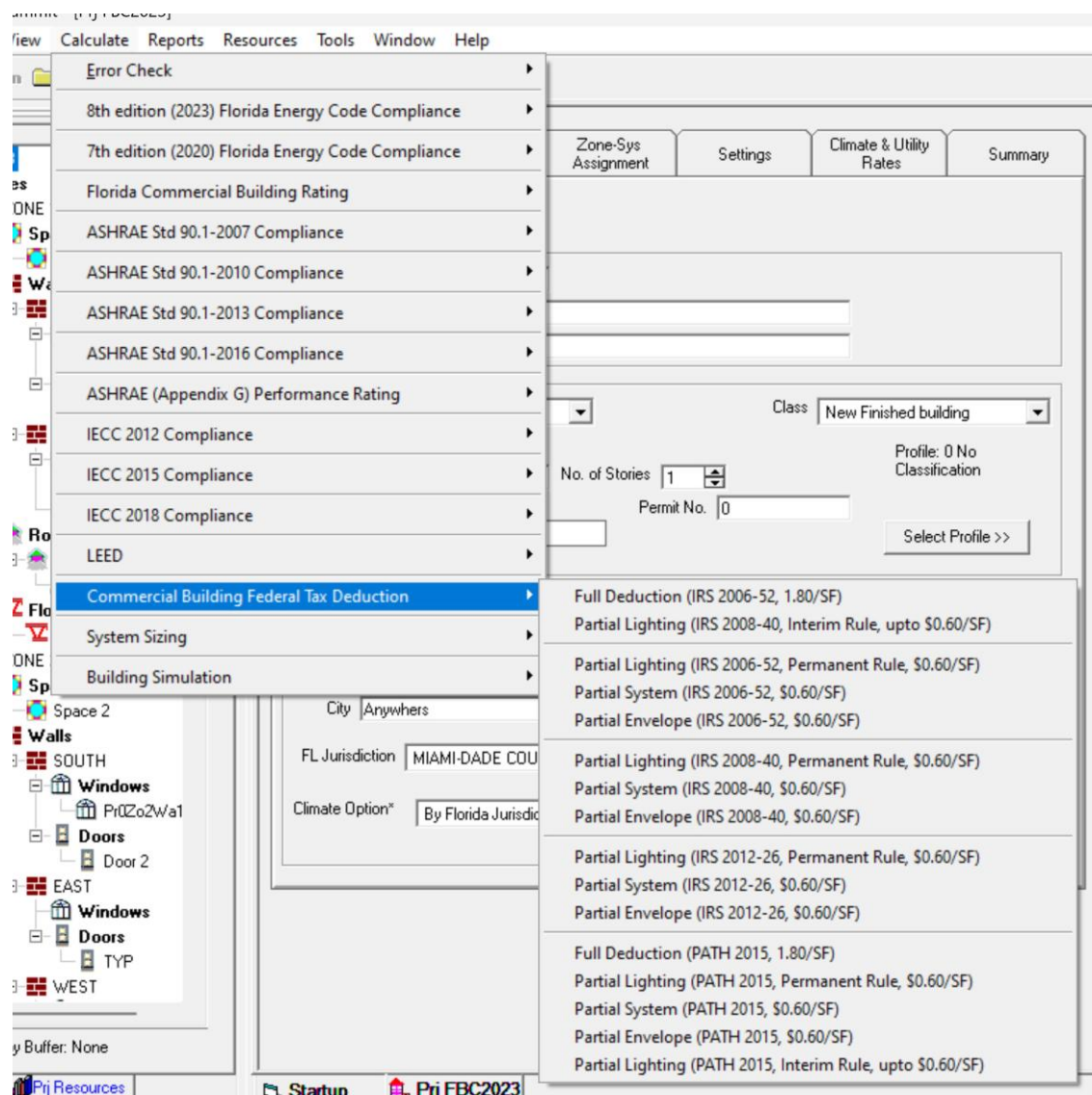
LEED

LEED represents The Leadership in Energy and Environmental Design (LEED) a US Green Building Council (USGBC) Rating System, the nationally accepted benchmark for the design, construction and operation of high performance green buildings. This menu option runs the following simulations.

- *LEED NC 2009* - This menu option will calculate the LEED EA Credit 1 points as per the LEED NC 2009 guidelines from the USGBC. EA Credit 1 points are awarded on the percentage difference between the annual energy use of the proposed versus the reference building constructed according to the ASHRAE

90.1 2007 Appendix G standards. The EnergyGauge software automatically calculates these USGBC points and displays a detailed report of the proposed versus reference energy use.

- **LEED NC 2009 - EA Credit 1 Report (Adobe not required)** - This option does the same calculation as the LEED NC 2009 menu option but displays the output report in a format that is required for the LEED NC 2009 EA Credit 1 PDF. This option does not require Adobe Acrobat Professional to be installed on the client computer. The user has the ability to print this report as a 'Microsoft XPS Document Writer' document and copy all the outputs in to the LEED PDF for submittal. The 'XPS' document can only be opened using Microsoft Internet Explorer.
- **LEED NC V4 - EA Credit 1 Report (w/o Online Filing)** - Runs the LEED New Construction (NC) v4 simulations that reports the number of credit points in the LEED rating system that a building can earn for improved energy performance. The EnergyGauge software automatically calculates these USGBC points and displays a detailed report of the proposed versus reference energy use.



2005 Commercial Federal Tax Deduction

Provisions in the Energy Policy Act 2005 allow for a tax deduction for energy efficient commercial buildings that reduce annual energy and power consumption by 50% compared to the ASHRAE 2001 Standard 90.1. A special approval code must be provided to use this feature. Once this code is specified by the user under the Help menu, and verified by the software, it need not be entered again till the license is valid. Following are the options available.

** Full Deduction (IRS 2006-52, IRS 2008-40, PATH 2015, \$1.80/SF)* - The full deduction is applicable if the whole building calculation shows a 50% or greater efficiency than the corresponding reference building for the ASHRAE Standard 90.1. This deduction is a maximum of \$1.80 per square foot of the building.

** Partial Lighting (IRS 2008-40, PATH 2015, Interim Rule, up to \$0.60/SF)* - As per IRS guidelines documented in the notice IRS 2008-40, any commercial building with an improvement of over 25% in the lighting power density as compared to the ASHRAE 90.1 2001 requirements can qualify for a tax deduction. The deduction is scaled appropriately, as per IRS guidelines, in the 25% to 40% savings range and a deduction of \$0.60 /SF can be taken for savings over 40%.

** Partial Lighting Deduction (IRS 2006-52, IRS 2008-40, IRS 2012-26 Permanent Rule, PATH 2015, \$0.60/SF)* - The partial deduction is applicable if the code calculation shows a 50% or greater efficiency of the building lighting than the corresponding reference building lighting for the ASHRAE Standard 90.1. This deduction is a maximum of \$0.60 per square foot of the building.

** Partial System Deduction (IRS 2006-52, IRS 2008-40, IRS 2012-26, PATH 2015, \$0.60/SF)* - The partial deduction is applicable if the code calculation shows a 50% or greater efficiency of the building mechanical system than the corresponding reference building mechanical system for the ASHRAE Standard 90.1. This deduction is a maximum of \$0.60 per square foot of the building.

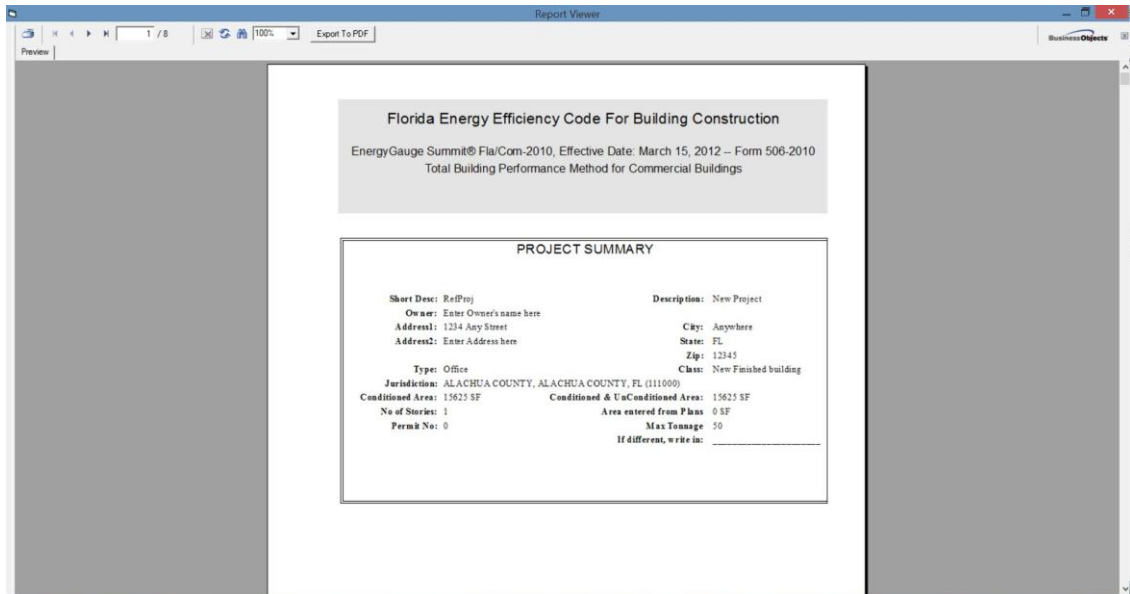
** Partial Envelope Deduction (IRS 2006-52, IRS 2008-40, IRS 2012-26, PATH 2015, \$0.60/SF)* - The partial deduction is applicable if the code calculation shows a 50% or greater efficiency of the building envelope than the corresponding reference building envelope for the ASHRAE Standard 90.1. This deduction is a maximum of \$0.60 per square foot of the building.

System Sizing - This is a beta feature that will size the user building using the DOE 2.1E simulation engine. The output report will provide the sized heating and cooling capacities for all system and plant components included in the building model.

Building Simulation

* *Run DOE2.1E Simulation* - This runs an energy (not code compliance) calculation on the design building using the DOE-2.1E simulation engine. It is not an option nor a requirement for code compliance. However, its output may be of useful to the designer.

**** Some of the 'Calculate' menu options may or may not be visible based on the version of the program purchased**

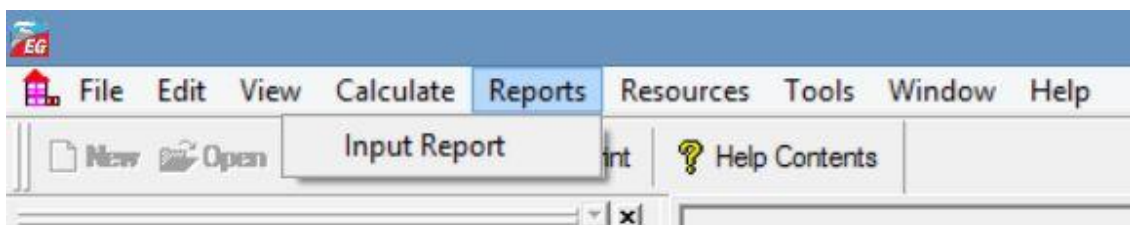


Report Viewer - After a calculation is run your report will show up in the report viewer. Use the arrow buttons to navigate through the various pages in the report.

* *Export to PDF* - Use the "Export to PDF" button to save a copy of the report in pdf format.



2.5 Reports Menu



Input Report - This produces a detailed report of the project inputs.

The screenshot displays the 'Project Information' tab of the EnergyGaugeSummit software. The interface includes several input fields and dropdown menus for project details. A 'Print Input Report' dialog box is currently open, asking the user if they want to include Profiles & Schedule data in the input report.

Project Information Tab:

- Description:**
 - Acronym: RefProj
 - Title: New Project
 - Owner: Enter Owner's name here
- Building Info:**
 - Type: Office
 - Class: New Finished building
 - Area From Plans: 0
 - No. of Stories: 1
 - Profile: 501 ACM Nonres
 - Rotate Building clockwise by (degrees): North (0 deg)
 - Permit No.: 0
 - Select Profile >>
- Location Info:**
 - Address: [Field]
 - City: [Field]
 - FL J: [Field]
 - Climate: [Field]

Print Input Report Dialog:

The Input Report now includes detailed data on Profiles & Schedules. This voluminous information need not be included in the Input Report for Florida Code Compliance.

Do you want to include Profiles & Schedule data in Input Report?

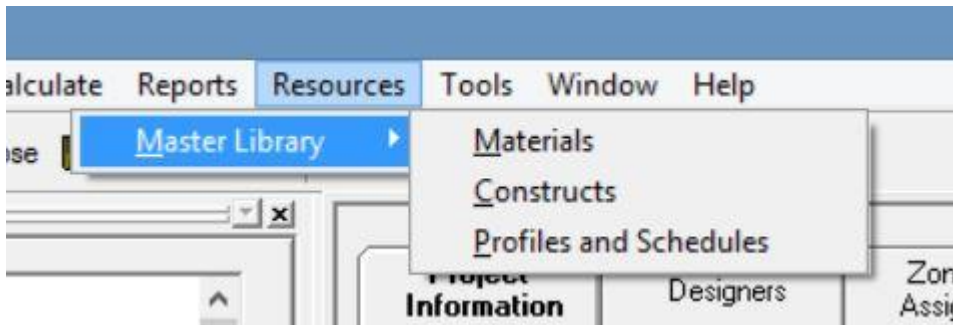
Buttons: Yes, No, Cancel

Before displaying the input report, the user is asked to select whether the Profiles & Schedules information should be included in the report. The Profiles & Schedules information is voluminous and is not required to be included for Florida Code Compliance but may be mandatory for other compliance calculations.

Note: In certain instances of renovation, addition or shell buildings, only some of the individual reports may be required. For example, if only a system is being replaced, only the system report may be required. Check your current energy code manual and local jurisdiction for specific requirements.



2.6 Resources Menu



Provides access to the following Master Library resources:

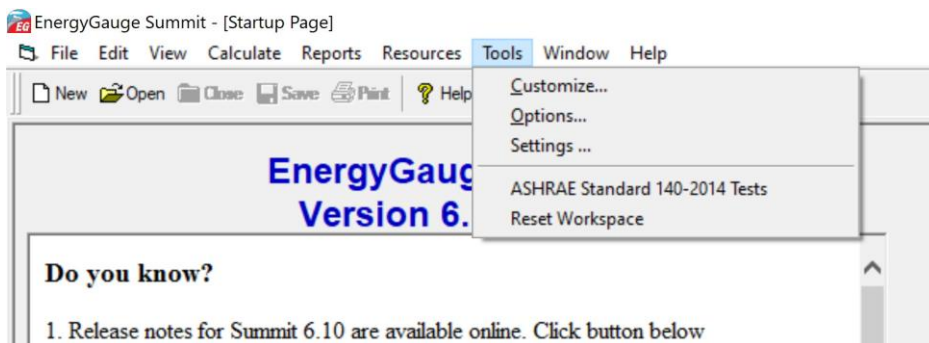
Materials - Displays the Master Library of Materials

Constructs - Displays the Master Library of Constructs

[Profiles and Schedules](#) - Displays the profiles and schedule data forms



2.7 Tools Menu



The Tools menu has the following sub menus:

Customize - This menu allows the user to customize the Tool bars.

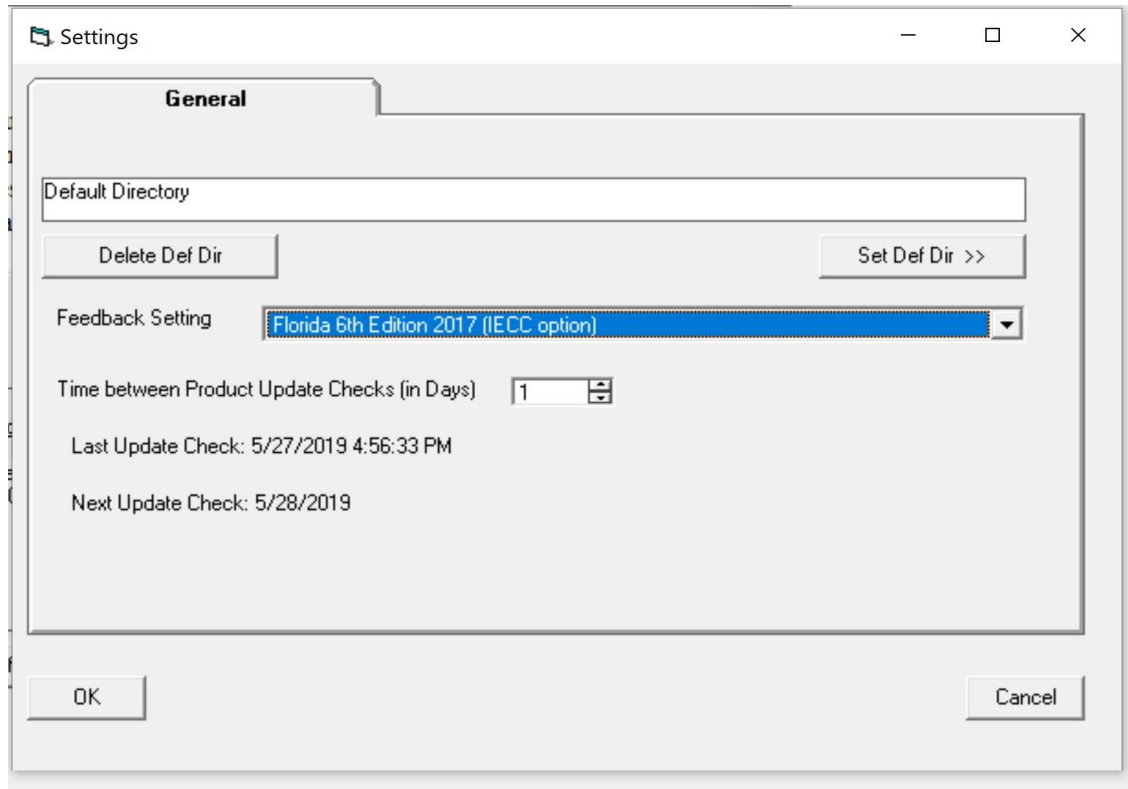
Options - This option allows the user to change the application user interface layout. Layout options include Tab location and style.

Settings - This allows the user to set, change or remove the default directory from which input files are initially opened.

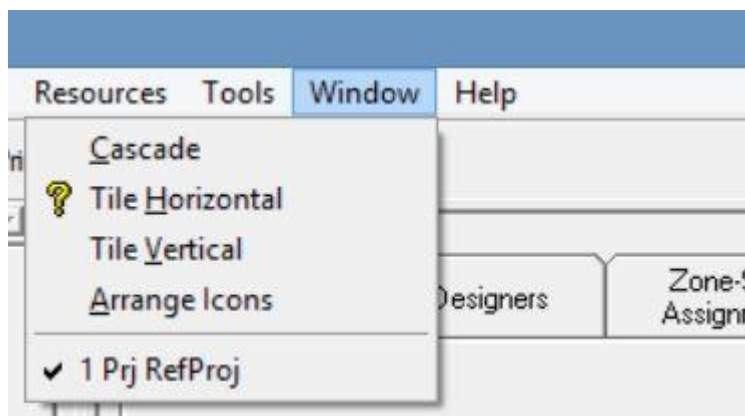
Another option in the setting menu is for the feedback control for users when entering building data. This feedback control when set to a particular form of calculation, for example should the user be running Florida 6th Edition 2017 IECC option, the feedback

dropdown menu should be set to Florida 6th Edition 2017 IECC option, thus allowing a user to see whether individual data fields fail or pass according to the rules of the feedback standard selected, when entering data in the forms. This setting is shown below:

Also, the interval, in number of days, between automatic update checks can be set here.



2.8 Window Menu



The sub menus under the Window menu provide the user with the option of viewing multiple forms in several formats.

Cascade - Reduces the size of open window and places them on top of each other

Tile Horizontal - All open windows are tiled horizontally.

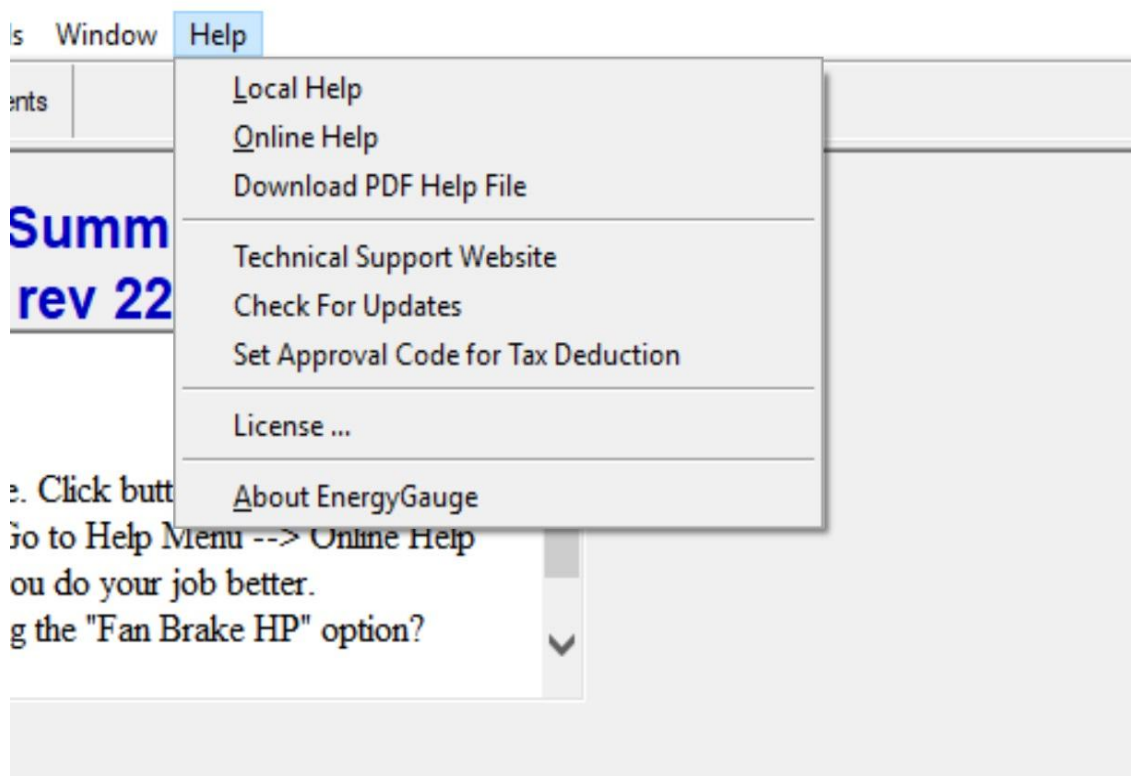
Tile Vertical - All open windows are tiled vertically.

Arrange Icons - The window icons are minimized and are arranged at the bottom of the screen.

Note: These menus are only for the visual convenience of the user. Window in this context does not refer to the 'Windows' or 'fenestrations'. in a building.



2.9 Help Menu



Sub Menus provides access to the Help system.

Local Help - This sub menu opens the contents dialog of the Help system. The user can navigate to any topic from the contents. In addition the user can print any topic displayed or book from the contents dialog.

EnergyGaugeSummit Help

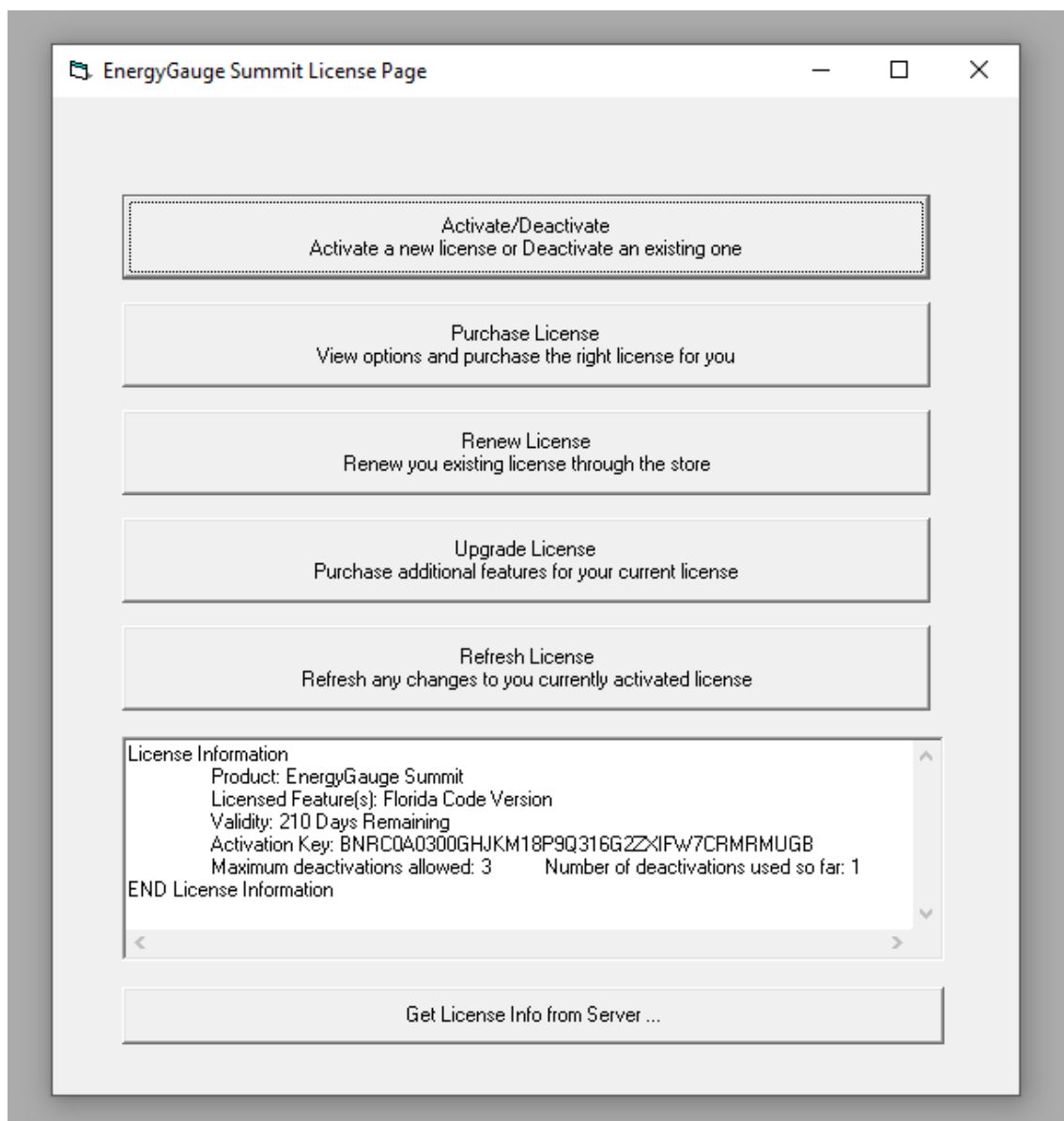
Online Help - This sub menu take you to the help system posted online with the latest available changes.

Download pdf Help file - This sub menu allows you to download and save locally the latest available help file in PDF form.

Technical Support Website - Directs the user to the EnergyGauge Summit technical support website.

Check for Updates - Checks online if any updates are available.

License ... - Takes you to the License action page, to activate deactivate etc. shown below



About EnergyGauge - Provides information about the installed version of EnergyGauge Summit.

Important Note: Pressing F1 key on your keyboard will provide context sensitive help.



3. Building Component Inputs

3.1 Project Data Input Form

Project Information Tab

Project Information	Designers	Zone-Sys Assignment	Settings	Climate & Utility Rates	Summary
Description Acronym RefProj Title New Project Owner Enter Owner's name here					
Building Info Type Office Class New Finished building Area From Plans 0 No. of Stories 1 Profile: 501 ACM Nonres Rotate Building clockwise by (degrees): North (0 deg) Permit No. 0 Select Profile >>					
Location Info Address 1234 Any Street Enter Address here City Anywhere State FL Zip 12345 FL Jurisdiction ALACHUA COUNTY ----- ALACHUA COUNTY Climate Option* By Florida Jurisdiction *Use "By Florida Jurisdiction" for Florida Compliance. Select other options here and go to "Climate..." tab					

Description

Acronym - A short description (up to 20 characters).

Title - A fuller description (up to 50 characters).

Owner - The building owner information.

Location Info

Address - The address where the building is located. Two lines are available for this.

City - The city where the building is located.

State - The state where the building is located

Zip - The zip code where the building is located.

Jurisdiction - The city and/or county with authority to issue your building permit.

Climate Option - For Florida code compliance, this option must always be set to "By Florida Jurisdiction".

Building Info

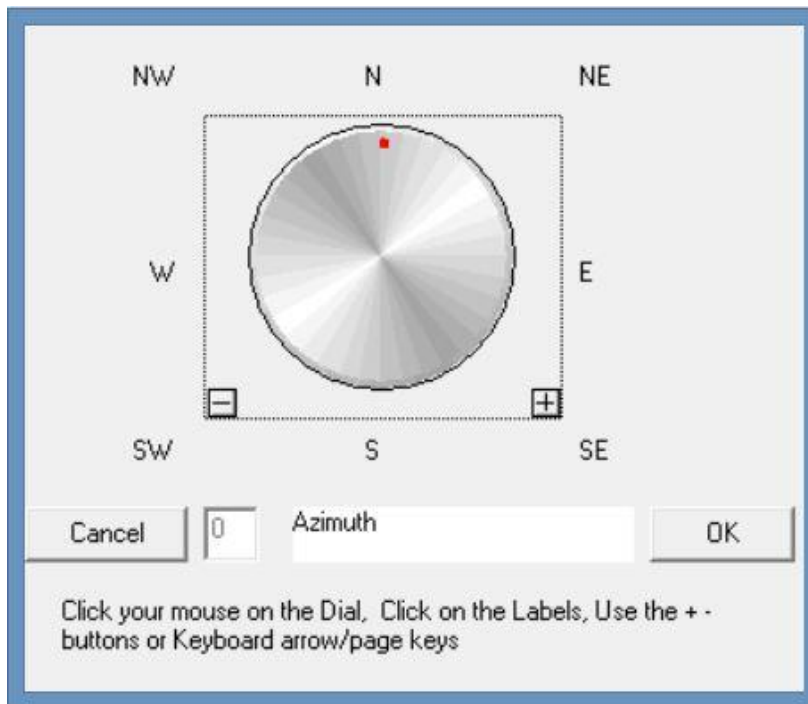
Type - The building classification based on use class from the [Type list](#).

Class - The class building such as New or existing for which compliance is being calculated.

Area from Plans - The total building square footage as indicated on the plans.

No of Stories - The number of the separate floors in this building project. The default is 1.

Rotate Building clockwise by (or Azimuth) - The orientation of the building is relative to true north. It is expressed in degrees from 0 to 360. The default is 0.0. Changing this angle has the effect of rotating the building about its z-axis (vertical axis). The default value is 0 and usually does not require modification. Clicking will open the Azimuth Dial window. Common values are: North = 0; South = 180; West = 270; East = 90.



There are 3 ways to enter the value

1. Click on the minus  or plus button  to select the value and then click on 'OK'.
2. Click on the E or W or N or S or NE or NW or SW or SE to enter the selected value and then click 'OK'.
3. Click on the wheel to select the desired value and then click 'OK'.

NOTE: Orientation of Walls, Roofs and Floors are relative to the building orientation. Any value other than zero for the building orientation will rotate all Walls, Roofs and Floors correspondingly during calculation.

Select Profile - Use the button to load a profile for the zone. The profile will include all the schedules for internal loads and gains in the building. For non-code purposes, the profiles and schedules can be entered and modified by the user. All code related simulations will use default values for profiles and building internal loads and gains schedules.

Designers Tab

Project Information	Designers	Zone-Sys Assignment	Settings	Climate & Utility Rates	Summary																								
<table border="1"> <thead> <tr> <th>Designers</th> <th>Name</th> <th>Registration/License No.</th> </tr> </thead> <tbody> <tr> <td>Prepared By</td> <td> John Doe </td> <td> 1234567 </td> </tr> <tr> <td>Owner/Agent</td> <td> </td> <td> </td> </tr> <tr> <td>Architect</td> <td> </td> <td> </td> </tr> <tr> <td>Electrical Designer</td> <td> </td> <td> </td> </tr> <tr> <td>Lighting Designer</td> <td> </td> <td> </td> </tr> <tr> <td>Mechanical Designer</td> <td> </td> <td> </td> </tr> <tr> <td>Plumbing Designer</td> <td> </td> <td> </td> </tr> </tbody> </table>						Designers	Name	Registration/License No.	Prepared By	John Doe	1234567	Owner/Agent			Architect			Electrical Designer			Lighting Designer			Mechanical Designer			Plumbing Designer		
Designers	Name	Registration/License No.																											
Prepared By	John Doe	1234567																											
Owner/Agent																													
Architect																													
Electrical Designer																													
Lighting Designer																													
Mechanical Designer																													
Plumbing Designer																													

Enter names of individuals and Registration as applicable. (up to 50 characters)

Zone-Sys Assignment Tab

Project Information	Designers	Zone-Sys Assignment	Settings	Climate & Utility Rates	Summary
System\ Zone	System: Pr0Sy2	System: Pr0Sy4			
Zone: Pr0Zo1	☒	☐			
Zone: Pr0Zo2	☐	☒			

For Method A Compliance all zones in the project must be assigned to systems by checking the appropriate box.

A system can have more than one zone assigned to it. However, multiple systems cannot be assigned to a single zone.

Settings Tab

Project Information	Designers	Zone-Sys Assignment	Settings	Climate & Utility Rates	Summary
IECC 2012 IECC 2012 Prescriptive Option: Additional Efficiency Package Option Energy Efficient HVAC Option (C406.2)					
☐ Ashrae 90.1-2010 Section 5.5.4.5 Exception? ☒ Use Building Area Method for Lighting Compliance ☐ Project Complies with IECC 2012 402.3.1.1 for increased Fenestration Area ☐ Project Complies with IECC 2012 402.3.1.2 for increased Skylight Area					
*Baseline Cooling Plant Capacity Multiplier 1 *Baseline Heating Plant Capacity Multiplier 1 Note: All Baseline System Capacity Multipliers have been moved to the System form under each system. The above multipliers apply only to the Baseline Plant					
☐ Override Internal Loads ☒ *Auto Size Proposed Building ☐ *Override Unmet Hours					
☐ **Draft Run ☐ **Use One Unit Per Zone					
* These entries are ignored for code compliance calculations. **Checking "Draft Run" performs Appendix G Baseline simulation in only one direction -- not all four ***Used one system per zone for rating calculations					

IECC 2012 Prescriptive Option - According to the IECC 2012 Code, a new building must pass the general given requirements for a new building, along with one of the energy efficiency packages described below. This dropdown menu allows the user to choose which efficiency package to include when running an IECC 2012 prescriptive method.

Energy Efficient HVAC Option - Uses the information from IECC 2012 c406.2 to determine if the minimum efficiencies of heating and cooling systems are within range with the requirements laid down in the same section.

Energy Efficient Lighting Option - Places stricter requirements on the lighting power density (LPD) of the whole building, as determined by the building type.

On-Site Renewable Energy #1 - Places the requirement that the sum total of all on-site renewables provide no less than 1.75 Btu (1850 W), or no less than 0.5 watts per square foot of conditioned floor space.

On-Site Renewable Energy #2 - Places the requirement that the sum total of all on-site renewables provide no less than 3 percent of the energy used within the building for mechanical and water heating equipment and lighting.

ASHRAE 90.1-2010 Section 5.5.4.5 Exception - Check or Uncheck depending upon if the building satisfies the exceptions outlined in ASHRAE 90.1-2010 section 5.5.4.5 concerning fenestration orientation.

Use Building Area Method for Lighting Compliance - If checked, compliance will be calculated using an overall building total LPD as opposed to a zone by zone compliance calculation.

Project Complies with IECC 2012 402.3.1.1 for Increased Fenestration Area - Check or Uncheck depending upon if the building satisfies the requirements for additional vertical fenestration area detailed in section C402.3.1.1 of the IECC 2012 codebook.

Project Complies with IECC 2012 402.3.1.2 for Increased Skylight Area - Check or Uncheck depending upon if the building satisfies the requirements for additional skylight area as detailed in section C402.3.1.2 of the IECC 2012 codebook.

Override Internal Loads - If selected, allows the user to modify the internal loads in the space from the 'Loads' tab on the 'Space Data Input Form'. Leaving this box unchecked will allow the program to use default values for the internal loads. Not applicable for the 2004 and 2008 Florida codes.

Baseline Cooling Plant Capacity Multiplier and Heating Plant Capacity Multiplier - If the unmet hours for the **baseline (or reference building)** needs adjusting to meet the unmet hour rule, (See ASHRAE 90.1 or 2014 Florida Code for details) these two parameters serve as capacity multipliers for the baseline building to adjust plant output. These parameters will not effect the unmet hours for the proposed building, these hours can be met by modifying the capacity of the proposed plant itself.

Auto Size Proposed Building - If checked, the proposed building will be internally auto-sized so that the unmet hours for heating and cooling are minimized. This should be used only when user-entered system capacity is unable to meet loads within bounds of unmet hours. This would happen if the schedules and internal loads are not well established for the proposed building and therefore the system capacity is unable to meet loads. First check your input system capacities before using this option.

Override Unmet Hours - If checked, this will completely override the unmet hour rule. **Do not use this until all other options and inputs have been verified.** Use of this option will require justification to the authority having jurisdiction.

Draft Run - Especially useful in saving time when running the LEED or ASHRAE Appendix G 2004 simulations. The user can quickly run a single orientation for the reference building instead of all 4 orientations to figure out how much of savings they

can get for the annual proposed energy use over the reference energy use. Once the acceptable level of percentage savings has been reached, this box can be unchecked to run all 4 orientations to get the final results. Reports generated when the '**Draft Run**' option is checked will have a note saying that it was a draft run and a full run needs to be completed for final report submission. Users should be aware that the single run in the '**Draft Run**' mode may not necessarily be the worst orientation for the reference building.

'**Use one unit per zone**' - Can be checked to help reduce the heating and cooling unmet hours if they are too high for the baseline building model for non-code (LEED, ASHRAE Appendix G) simulation runs. When the box is checked, and the proposed building model contains a single system assigned to multiple zones, each of those multiple zones will be assigned their own system in the baseline model. This will help reduce the unmet hours for the baseline run in the case where a single zone system is forced to serve multiple zones due to equipment selection as required by ASHRAE Appendix G.

Climate & Utility Rates Tab

Project Information Designers Zone-Sys Assignment Settings **Climate & Utility Rates** Summary

All these data not required for Florida Code Compliance

- + Alabama
- + Alaska
- + Arizona
- + Arkansas
- + California
- + Colorado
- + Connecticut
- + Delaware
- + District of Columbia
- + Florida

Select by Location Data

Latitude Longitude

Cooling Degree Days Heating Degree Days

Summer Design Temp Winter Design Temp

Altitude

Choose County, if listed here: Weather Date Set

Climate Location: ALACHUA COUNTY, ALACHUA COUNTY, FL
 Weather file: C:\Program Files (x86)\Common Files\EnergyGauge\Weather\USA\TMY3\FL_GAINESVILLE_REGIONAL_AP.tm3

Utility Rates ☐ Use these Rates for 2010 Florida Code Instead of Defaults

	Fuel	Fuel Type	Description	Monthly Charge	Energy Charge (\$)	Energy Charge Unit	Demand Charge (\$/Unit)	TOU Schedule
1	1	Electricity	Res1	\$ 0.0000	\$ 0.1000	kWh	\$ 0.0000	None
2	2	Natural Gas	Res2	\$ 0.0000	\$ 0.5000	therm		None
3	3	Fuel oil	Res3	\$ 0.0000	\$ 1.0000	gal		None

Allows the user to choose the climate location for the design building and to enter local utility rates for fuel charges.

TMY3 (Typical Metrological Year) weather data files are now used by default for all code simulation runs in EnergyGauge Summit. For all non-code simulation runs (LEED, ASHRAE Appendix G etc.), the user has the option to use the older TMY2 weather data set. TMY3 weather data sets are derived from more accurate and more recent weather data and are recommended to be used for greater simulation accuracy. These data sets are now available for over 1000 locations in the United States.

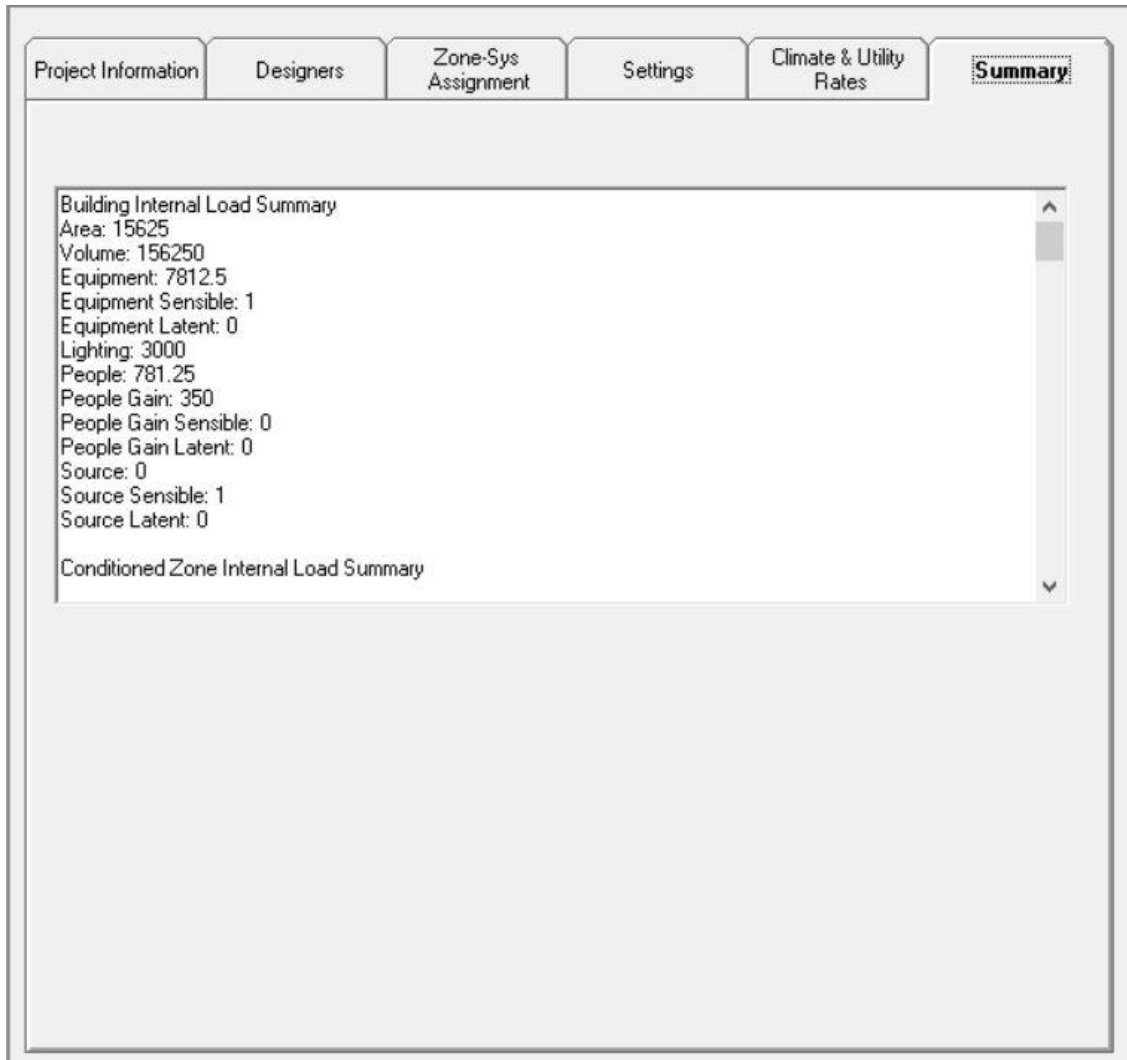
To use time-of-use rates, first check the “Use TOU Schedule?” column. Clicking the button on the “TOU Schedule” column will lead the user to the TOU schedule used. The user may choose to use another TOU schedule or create one under the main menu Edit -> Profiles and Schedules, and then invoke the newly created schedule here. Note that TOU schedules are applicable only for electric rates and not for other fuels.

☐ Use these Rates for 2010 Florida Code Instead of Defaults

If checked, this entry tells the program to use user entered utility rates and time of use rates instead of internal default rates for

the 2010 Florida code compliance. This entry has no effect on 2008 and 2004 Florida code compliance and ratings. For other calculations always the user entered rates are used.

Summary Tab



Project Information	Designers	Zone-Sys Assignment	Settings	Climate & Utility Rates	Summary
Building Internal Load Summary Area: 15625 Volume: 156250 Equipment: 7812.5 Equipment Sensible: 1 Equipment Latent: 0 Lighting: 3000 People: 781.25 People Gain: 350 People Gain Sensible: 0 People Gain Latent: 0 Source: 0 Source Sensible: 1 Source Latent: 0 Conditioned Zone Internal Load Summary					

The summary window presents all the information entered by the user for the particular project.

[Related Topics](#)



3.2 Zone Data Input Form

Zone Load Data Tab

Zone Load Data		Summary
Acronym	Desc	Type
Pr0Zo1	Zone 1	Conditioned Zone
Select Category >>>	Classroom/Lecture Hall	
☐ Is enclosed (per ASHRAE 90.1-2010 Section 5.5.4.2.3) ☐ IECC 2012 402.3.1.1 Exception for increased Fenestration Area		☐ Use Building Load Profile OR Select Profile >> Selected Profile: 501 ACM Nonres ☐ IECC 2012 402.3.1.2 Exception for increased Skylight Area
Dimensions Total Space Area = 10000.0 SF Total Space Volume = 100000.0 CF Effective Zone Height = 10.0 ft Area not included in Spaces (SF) 0 Number of Zones like this one 1		Equipment Total: = 5000.0 W Sen Frac: = 1.0 Lat Frac: = 0.0
Lighting Lighting Load from Spaces = 1500.0 W .. Type Compact Fluorescent ☐ ASHRAE 90.1 2010 #5.5.4.2.3 Exception c or d Primary Side Lighted Area (Per ASHRAE 90.1 2010 #5.5.4.2.3 Exception) 0 Secondary Side Lighted Area (Per ASHRAE 90.1 2010 #5.5.4.2.3) 0		Thermostat Type Proportional Range 4 Occupants Number of people = 500.0 Source Electric Source Tot: = 0.0 Btu/hr Sen Frac: = 1.0 Lat Frac: = 0.0

Acronym - A short description (up to 20 characters).

Description - Enter a detailed description (up to 50 characters).

Type - Specifies the type of zone

Conditioned - maintains zone air at desired conditions.

Unconditioned - Zone is not conditioned.

Plenum - Zones that act as conduits for return air.

Zone Category - Select the category of the zone from a pre-defined list. This is required as for example, the building type may be an assembly, but may have a zone that serves as an office located in it. The building type selected on the project form will then be an assembly, but the category selected for this zone will be of type 'Office'.

Is enclosed (per ASHRAE 90.1-2010 Section 5.5.4.2.3) - If checked, will classify the zone as enclosed as per the description given in ASHRAE 90.1-2010 5.5.4.2.3

IECC 2012 402.3.1.1 Exception for increased Fenestration Area - If checked, will allow for calculations to be run with an increased allowance for vertical fenestration. To check if such an exception is allowed, refer to the IECC 2012 code.

IECC 2012 402.3.1.2 Exception for increased Skylight Area - If checked, will allow for calculations to be run with an increased allowance for skylights. To check if such an exception is allowed, refer to the IECC 2012 code.

Use Building Load Profile or  - Allows to user to apply a pre-existing profile to the zone, or use the overall building profile when attempting to run load calculations.

Dimensions

NOTE: The area and volume of the zone are automatically calculated by summing the individual values for the spaces under the zone. This occurs any time a change is made to space dimensions or spaces are added or deleted. There is, however, a limitation on the per Zone Area and Volume thus calculated. Maximum Zone area is limited to 100,000 SF and maximum Zone Volume is limited to 1,000,000 CF. However, these can be overcome by defining additional Zones or using the Zone Multiplier for multiple zones that are exactly the same.

Area not included in spaces [SF] - Represents extraneous area that has not been included under spaces. This is added to the total space area under the zone. If none, set to 0.

Number of Zones - Specifies the number of identical zones in the building to the current one. If two or more zones are identical, use this entry rather than creating additional zones.

Thermostat

The thermostat will control the zone to the particular set-point as specified in the zone set point schedule for heating or cooling. It will be responsible for turning the HVAC system on or off based on zone heating or cooling load requirements. The user should input the type of thermostat (eg. proportional etc.) and the range or the dead band in which the thermostat should control the zone temperature to.

Lighting

Type - Specifies the predominant type of overhead lighting used in the zone. Note that lighting power and type is specified in the space forms.

ASHRAE 90.1-2010 5.5.4.2.3 Exception C or D - If checked, will allow for a calculation to run with reduced daylight area under skylights. This reduced area is based upon the amount of primary and secondary sidelighted area.

Primary Sidelighted Area - The area in square feet of primary sidelighted area with an effective aperture greater than 0.15 and with general lighting controlled as described in section 9.4.1.4 of ASHRAE 90.1-2010.

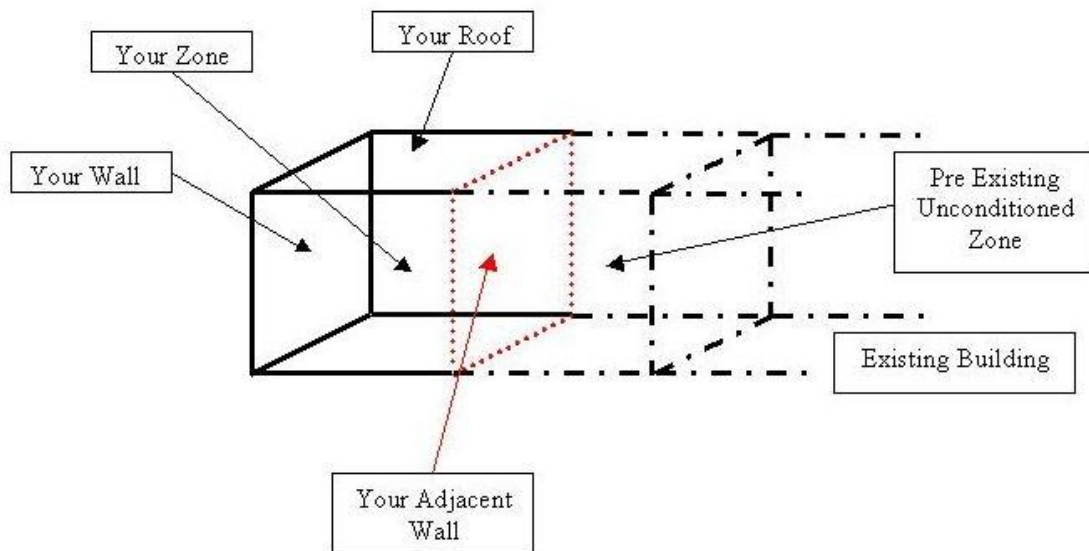
Secondary Sidelighted Area - The area in square feet of secondary sidelighted area with an effective aperture greater than 0.30 and with general lighting controlled by continuous daylight dimming.

Source

Specifies internal source type due to sources other than people, lights, or equipment. Note that the source rates rate is specified in the space forms.

HOW TO HANDLE A PRE-EXISTING UNCONDITIONED ZONE THAT IS NOT PART OF YOUR DESIGN BUT HAS YOUR ENVELOPE NEXT TO IT

In some instances, you may have to account for a zone that pre-exists and is not part of your design. Since the calculation requires a physical zone whenever an adjacent envelope is specified next to it, the following procedure suggests how to accomplish it with minimal input and effort.



Steps to account for an Zone that pre-exists.

- 1) Add a Zone. You may choose to label it as '**PRE-EXIST**'.
- 2) Set the Zone to '**Unconditioned**'.
- 3) Add a Space under this new Zone.
- 4) Set Space category to '**Unlisted**'.

- 5) Set the width and height of the space equal to the width and height, respectively, of your adjacent wall.
- 6) Set the depth of the space to a minimum value, say 1 ft.
- 7) Set No. of Task locations to zero.
- 8) Add a lighting to the space. Use minimum Watts (equal to what is indicated by the Budget, or say 5 Watts).
- 9) Set Control Type to '**Security Lighting**'.
- 10) Set No. of Controls to Zero.
- 11) You can now set your adjacent wall next to this new '**PRE-EXIST**' zone.
- 12) If you are running Method A, you will have to assign the '**PRE-EXIST**' zone to some system in the project. As long the '**PRE-EXIST**' zone is set to '**Unconditioned**', there will be no energy impact.

Zone Summary

Zone Load Data

Summary

Zone Air

Wall Summary

Exterior Walls: 1505.28 (1505.28 Gross)

Adjacent Walls: 0 (0 Gross)

Underground Walls: 0 (0 Gross)

All Walls: 1505.28 (1505.28 Gross)

Window Summary

Exterior Windows: 325 (325 Gross)

All Windows: 325 (325 Gross)

Roof Summary

Exterior: 906.304 (906.304 Gross)

[Related Topics](#)



3.3 Space Data Input Form

Space Data Tab

The screenshot shows the 'Space' tab in the EnergyGaugeSummit software. The interface includes a top menu bar with 'Print' and 'Help Contents' options. The main window is divided into three tabs: 'Space', 'Lighting', and 'Internal Loads'. The 'Space' tab is currently selected and contains the following elements:

- Acronym:** A text field containing 'Space 1'.
- Description:** A text field containing 'Space 1'.
- Select Space Category >>>:** A button to open a category selection dialog.
- Dimensions:** A section containing input fields for 'Width' (50), 'Depth' (50), 'Height' (23), and 'Multiplier' (1). Below these fields, the calculated 'Area = 2500 (2500 Gross)' and 'Volume = 57500 (57500 Gross)' are displayed.
- Footnote:** A small text note stating '*RCH: Room Cavity Height. See ASHRAE 90.1 2010/13 #9.6 for definition'.

A prominent red text overlay in the center of the 'Space' tab reads: 'Lighting inputs have been moved to the next tab titled "Lighting"'. This indicates that the 'Lighting' tab contains the relevant input fields for lighting calculations.

Acronym - A short description (up to 20 characters).

Description - A detailed description (up to 50 characters).

Category - Select a space type from the list. For space type not listed select the one most representative from the list. This is a key selection that will affect the lighting allowance.

Dimensions

NOTE: The area and volume of the zone to which the space belongs are automatically calculated by summing the individual values for all spaces under the zone. This occurs any time a change is made to space dimensions or spaces are added or deleted. Maximum Zone area is limited to 100,000 SF and maximum Zone Volume is limited to 1,000,000 CF. Space dimensions specified here should conform to this limitation.

Width - Specify the space width in feet.

Depth - Specify the space depth in feet.

Height - Specify the space height in feet.

Multiplier - Used to specify the number of identical spaces. Use this input for spaces that are exactly the same, instead of adding a new space under the zone.

RCH - Specifies the Room Cavity Height. This is defined to be the distance between the workplane and the luminaire mounting height of the space.

Lighting

Space
Lighting
Internal Loads

Lighting

Additional Control Method (See ASHRAE 90.1-2010/13 Table 9.6.2/3)
Manual, continuous dimming control or Programmable multi-level dimming control

Manual, continuous dimming control or Programmable multi-level dimming control

Space to which this Additional Interior Lighting allowance be allotted
Zone: ZONE 1-Space: Space 1

Lighting Total = 750 (750 Gross)

☐ Has Primary Side Daylighting

☐ Has Top Daylighting

Note there is a change in inputs for lighting controls.
Use input column "FBC 8thEd ..." for FBC 8th Edition, ASHRAE 2019, IECC 2021 and beyond
Use input columns "All Other Codes ..." for prior codes

☒ Hide Pre-8th Ed Columns

	Type	Category	No. of Luminaires	Watts per Luminaire	FBC 8thEd Control Type	FBC 8thEd Control Exception	Area Lighted
1	Compact Fluorescent	General	15	50.00	[1][7][8]	[4]	

Enter below justification for any Control Exception taken, such as, code section etc

section 405.22 page 223

Compliance Feedback set to: Florida 8th Edition 2023 (IECC option)
 Lighting Power Compliance PASSES
 Lighting Power Design: 750 Effective: 750 Allowance: 761.2953 Compliance:
 Lighting Controls FAILS
 Control type 3 required for compliance not found
 At least one additional control type [9] required for compliance
 Required Controls All of [3]; one of [9]
 Note: Lighting Power Compliance is based on entire building

Additional Control Methods - Allows the use of additional space-wide lighting controls. These are meant to allow for additional interior lighting power allowances, as per ASHRAE 90.1-2010 Table 9.6.2.

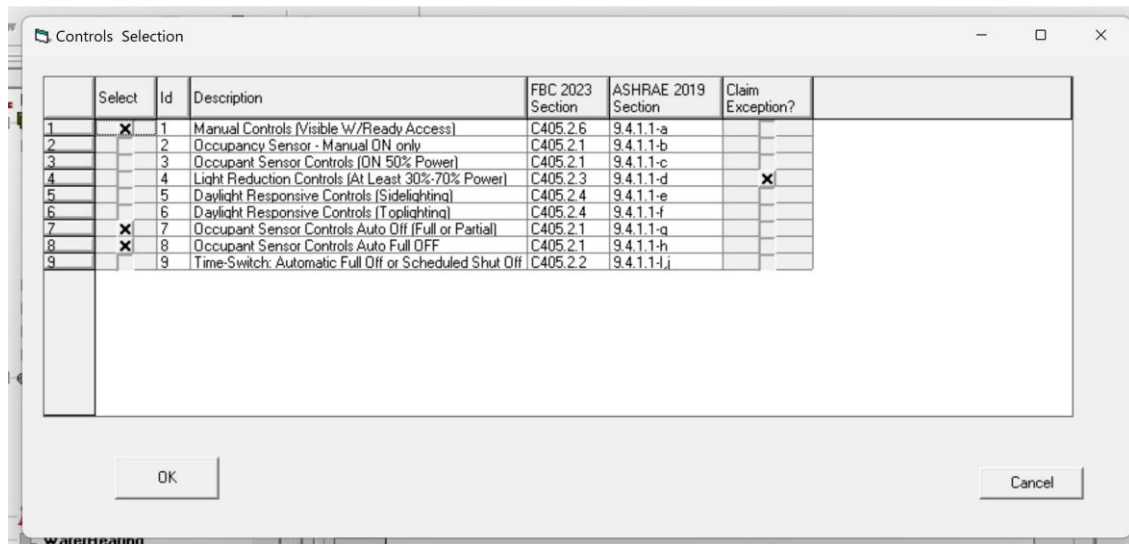
Type - This field describes the various lighting types. They are available in the drop down box. Click here for [details](#).

Category - Classified the use of the lighting.

No. of Luminaires - Specify the number of light fittings in that space.

Watts per Luminaire - Specify the lighting wattage per fitting.

Control type - By clicking on the cell button user can choose an number of control types and exemptions for FBC2023 & beyond -- as shown below



No. of Controls - This field shows the number of control points for the particular lighting. - pre 2023 FL Code

Area Lighted - For a few categories of the lighting type (for example: category 22 - Specific Retail Display), this extra field requires the user to enter the area lighted. This field is automatically unlocked for the categories for which this values is required.

See the section on [Working With Grids](#) to add, delete and modify the Lighting data.

Loads Tab

Space	Lighting	Internal Loads
Equipment W/SqF 0.5 Kw 0 Total Sensible 1 Latent 0	Lighting Loads W/SqFt 1.3 Light KW 0 Task Lt W/SqFt 0 Task Lt KW 0 Source Btu/Hr 60	Occupants Number of People 0 Area/Person 100 Heat Gain (Total) 350 Heat Gain (Sensible) 0 Heat Gain (Latent) 0 Occp.Load Outdoor Air CFM Per Person 15 CFM per SF 0

'Equipment' loads will impact zone 'heating/cooling' loads.

'Source' loads will be counted as miscellaneous (electric) energy use ('Process Loads' for LEED and other Rating calculations).

Note: User specified loads will not apply unless the 'Override Internal Loads' box is checked in the 'Summary and Settings' tab of the main Project interface.

For code calculations, the Loads tab on the Space form is provided for information only. Default values are set by the program based on the space type selected. All inputs on the space loads form will be used in case of non-code simulation (LEED, Federal Tax Deduction Calculations etc.) only when the 'Override Internal Loads' box on the 'Summary&Settings' tab is checked.

Equipment - Equipment loads should be entered in either 'W/sq ft' or 'kW'. If both values are entered, they are added together for a combined load in each space. All the space equipment loads are summed up as the total equipment load on every zone. Equipment load is purely sensible in nature and contributes to the heating or cooling loads in the zone. The 'Equipment' schedule from the associated Zone Profile will control the amount of equipment load on the zone for every hour in the annual simulation. The equipment load is typically the thermal (heat load) fraction of all the plug loads in that space.

Source - The source load is entered in Btu/hr and is the sum of all the plug loads in the space. Source loads do not contribute to the heating or cooling loads in the space and zone.

Occupants - The maximum number of 'Occupants' in the space should be entered as either a value of 'Number of People' or as a value of 'Area per person' or as a combination of the two. The people load on the space will be modified as per the 'People' schedule specified in the profile associated with the zone that contains the space. Occupant load will affect the outside (fresh) air fraction required for the zone during an annual simulation unless a fixed value of outside air has been specified at the zone level

Outdoor Air: Note that outdoor air inputs have been moved from Zone form to Space form.

[Related Topics](#)



3.4 Wall Data Input Form

Wall Info

Wall Info

Wall Info

Acronym

Description

Select Construct >> ApLbCo1102 - Construct 1102 (U=0.5800 R=1.7241 Den=46.9440 HCapPerSF=18.6997 Abs=0.7000 Emm=0.9 Simple Construct)

Location: Sel Adj Zone >>

ASHRAE Category:

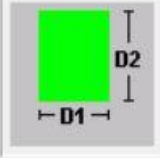
☐ Is Store Front Per ASHRAE 90.1-2010 5.5.4.4.1 (c)

Dimensions

Shape Type:

Dimension1: Multiplier:

Dimension2:



Tilt Orientation

Area: 1000 Total Area: 1000

Window Area incl Frame: 80 per Unit; (80 Gross)

Door Area : 28 per Unit; (28 Gross)

Misc

Gnd Refl Inside Vis Refl Outside Emiss

Infl Coef Inside Sol Abs

Acronym - A short description (up to 20 characters)

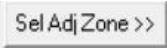
Description - A detailed description (up to 50 characters)

Select Construct - Allows selection of the construct (sometimes called wall type or envelope type) for the wall. Clicking this button opens a list of all constructs in the Project. You can then select a construct for the wall from the list. If a suitable construct is not available in the list, you must first add (or build) the construct in Project Resources before being able to select that construct for the wall.

Location - The explanation given here applies to Walls, Roofs and Floors.

Exterior - Implies that the surface is exposed to the outside (eg. Exterior Walls, Exterior Roofs, Floor that are raised above grade and exposed to the outside).

Underground - Implies that the surface is on or below ground (eg. basement walls, on-grade and below grade floors).

Adjacent to another zone - Implies that the surface is adjacent to another previously defined zone (eg. partition walls and ceiling between Stories). In this, case the zone to which this is adjacent to must be selected using the selection button 

The note below applies ONLY to version 1.21 and below

Important Note: The Adjacent zone selected here must always be a previously defined zone in the project hierarchy. It cannot be a succeeding zone in the project hierarchy. So the first zone in the project cannot have adjacent surfaces defined. But the second zone in the project can have surfaces that is adjacent to the first zone. However it cannot have surfaces defined that are adjacent to a third zone which comes after this one in the hierarchy.

Beginning Version 1.22, the above restriction does not apply and the user is not required to order the zones as above. The program automatically attempts to reorder zones as required and issues an error message when circular references in adjacent envelopes are discovered. A circular reference is one in which, for example, an envelope in Zone A is adjacent to Zone B, and another envelope in Zone B is adjacent to Zone A.

ASHRAE Category - Allows the user to select the appropriate ASHRAE category for the building component. This determines the maximum allowable assembly U-value for that component in the baseline (reference) building

Is Store Front Per ASHRAE 90.1-2010 5.5.4.4.1(c) - Check or Uncheck if this wall serves as a store front, as layed out by ASHRAE 90.1-2010 Section 5.5.4.4.1 (c). This will allow for the fenestration area on this wall to be ignored when determine the total vertical fenestration area percentage fo the building.

Dimensions

Shape Type - There are various shape types that can be entered. Note that the required number of dimensions will become active and should be entered upon selecting a shape type. The various shape types are:

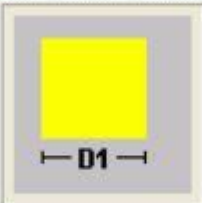
Square

Dimensions

Shape Type: Square

Dimension1: 100 Multiplier: 1

Tilt 90 Orientation North (0 deg)



Rectangle

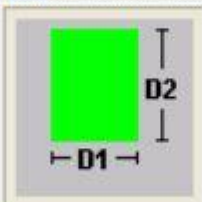
Dimensions

Shape Type: Rectangle

Dimension1: 100 Multiplier: 1

Dimension2: 10

Tilt 90 Orientation North (0 deg)



Triangle

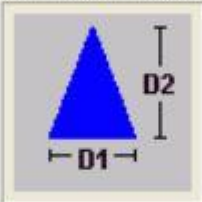
Dimensions

Shape Type: Triangle

Dimension1: 100 Multiplier: 1

Dimension2: 10

Tilt 90 Orientation North (0 deg)



Semicircle

Dimensions

Shape Type: Semicircle

Dimension1: Multiplier:



Tilt Orientation

Trapezoid

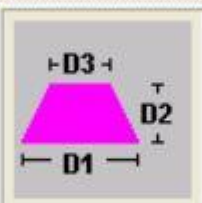
Dimensions

Shape Type: Trapezoid

Dimension1: Multiplier:

Dimension2:

Dimension3:



Tilt Orientation

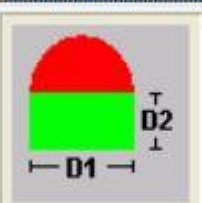
Rectangle with Semicircle

Dimensions

Shape Type: Rectangle with Semicircle

Dimension1: Multiplier:

Dimension2:



Tilt Orientation

Rectangle with Triangle

Dimensions

Shape Type: Rectangle with Triangle

Dimension1:

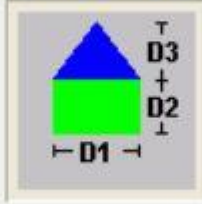
Dimension2:

Dimension3:

Multiplier:

Tilt:

Orientation: North (0 deg)



If the shape selected is a *Rectangle with Triangle* or *Trapezoid* then the user has to enter **3 dimensions** in the dimension boxes.

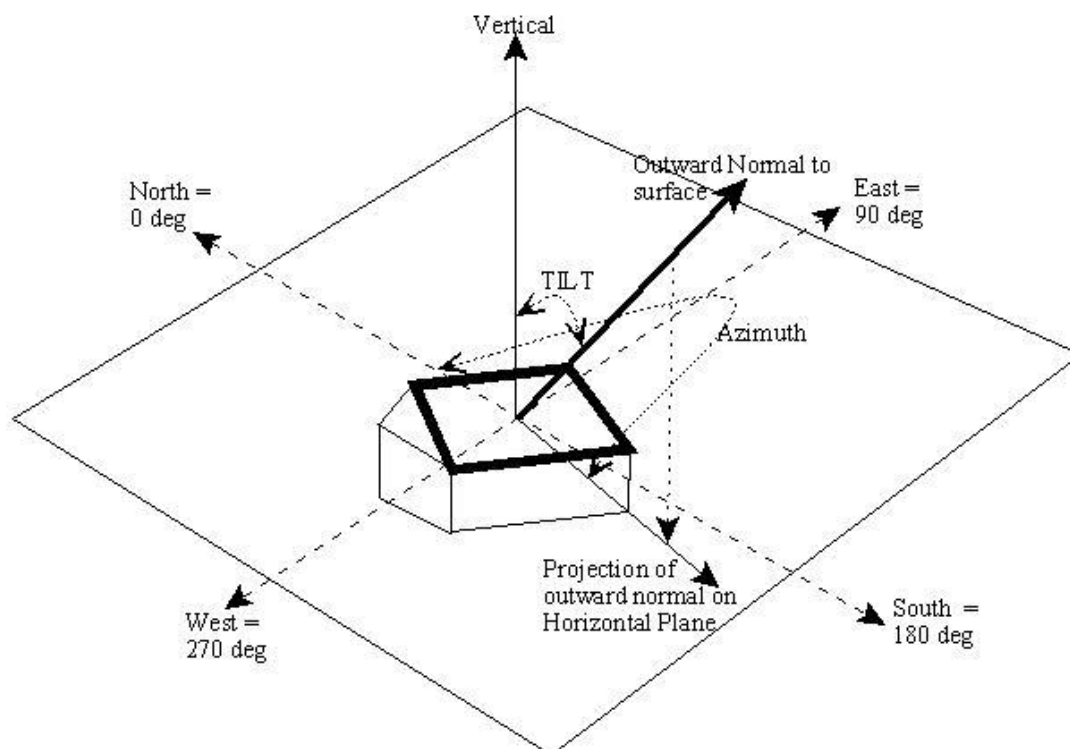
If the shape selected is a *Square* then the user has to enter **1 dimension**.

For rest of the shapes only **2 dimensions** have to be specified.

Dimensions [ft] - The required dimensions based on the particular shape selected.

Multiplier - Used to specify the total number of identical wall panels located in the same plane. If two or more walls are identical, use this entry rather than adding more walls to your project.

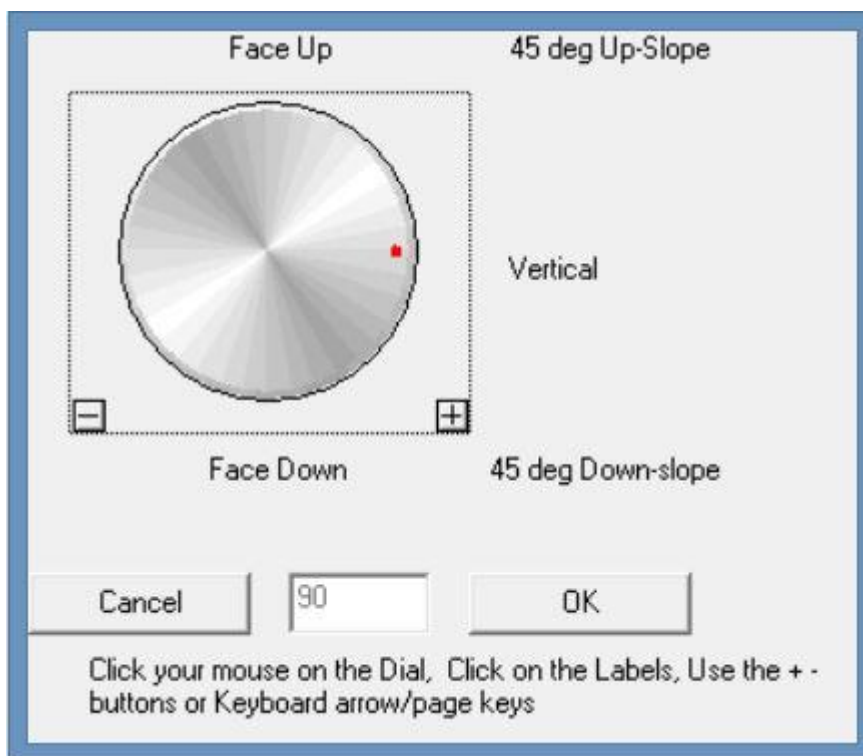
Placement



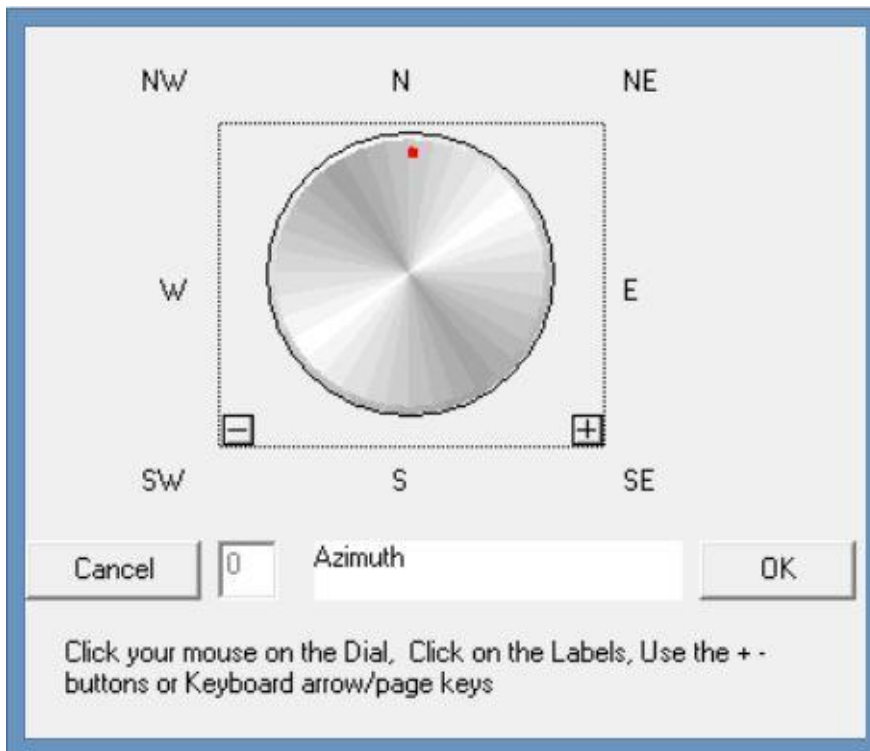
The figure above gives the general principles of determining the Tilt and Azimuth angles. An outward normal is first drawn from the exterior side of the surface. An outward normal is a line that is perpendicular to the plane of the surface in question. The tilt angle is the angle made by the outward normal with the vertical axis as shown in figure.

The azimuth is the direction the surfaces faces. For vertical surfaces, it is the direction the outward normal faces. To obtain the direction for a non vertical surface, first project the outward normal to the horizontal plane. The direction the projected line faces is the Azimuth, with North = 0, East = 90, South = 180 and West = 270.

Tilt - Surface tilt. Vertical = 90 (eg. Vertical Wall); Horizontal facing up = 0 (eg. Flat roof); Horizontal facing down = 180 (eg. floor). [Click](#) cell to set value. The following dialog opens:



Orientation (Azimuth) - Surface orientation (North = 0, South = 180, West = 270, East = 90). [Click](#) cell to set value. The following dialog opens:



NOTE: Orientation of Walls, Roofs and Floors are relative to the building orientation. Any value other than zero for the building orientation will rotate all Walls, Roofs and Floors correspondingly during calculation.

Miscellaneous

The following are not user editable and are provided for information only. The defaults are set by the Application.

Gnd Refl - Ground Reflectance is the solar reflectance of the ground.

Infl Coef - Infiltration Coefficient is the infiltration flow coefficient used to compute the infiltration resulting from cracks in a wall.

Outside Emiss - Outside Emissivity is the Emissivity of the exterior side of the wall.

Inside Vis Refl - Inside Visible Reflection is the fraction of light reflected by the interior side of the surface.

Inside Sol Abs - Inside Solar Absorbance is the fraction of radiation absorbed by the interior side of the surface. Radiation may fall on an interior surface through fenestration placed on other surfaces (eg. Sunshine falling on floors through a window or skylight).

[Related Topics](#)



3.5 Window Data Input Form

Window Info

Window Info

Acronym:

Desc:

Select Glass Type>>

Desc: ApLbWnd278 ---
Glass type: User Defined Panes=1 SHGC=0.39 U=0.67 VLT=1

☐ Fixed (Otherwise operable) Usage:

SetBack: Inside Vis Refl: Infil Coef:

Internally shaded ☐ with:

☐ IECC2012 402.3.3.2 Increased SHGC

Dimensions

Shape: Multiplier:

Dimension1: Dimension2:

Area: 80

Total Area: 80

☒ Shading Projection Factor = 0.000

Shading Data

	Dimension A	Dimension B	Dimension W	Dimension H	Dimension D	Angle
Overhang	0	0	0		0	90
Left Fin	0	0		0	0	
Right Fin	0	0		0	0	

Acronym - A short description (up to 20 characters)

Description - A detailed description (up to 50 characters)

Select Glass Type - Select the fenestration type for the window. Clicking this button opens a list of all fenestration in the Project. User can then select a fenestration for the window from the list. If a suitable fenestration is not available in the list, you must first add (or build) the fenestration to your Project Resources before being able to select that fenestration for the window.

Usage - Selects window usage type.

SetBack - Distance that the window is recessed into the wall in feet.

Inside Vis Refl - Inside Visible Reflection is the amount of light reflected from the interior side of the window. Not user editable.

Infl Coef - Infiltration Coefficient specifies an infiltration flow coefficient used to compute the infiltration resulting from cracks in the window. Not user editable.

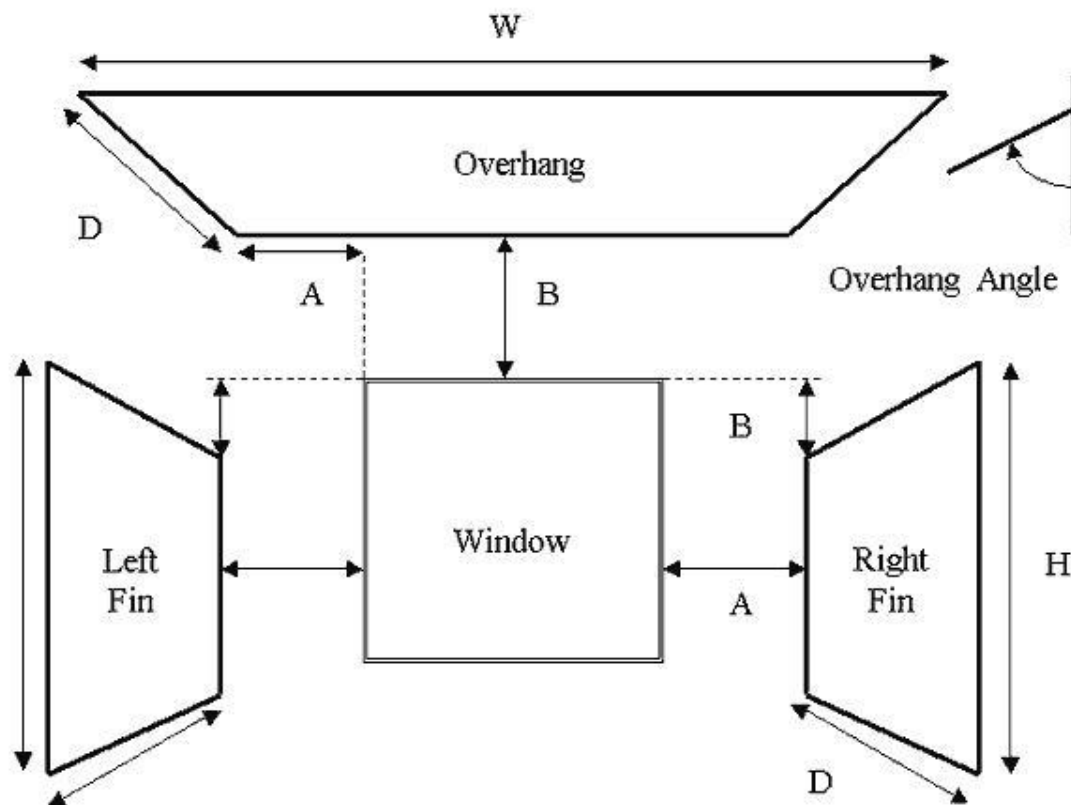
Internal Shading - Checking the box for Internal Shading will allow the user to specify the type of internal shading from a drop down box.

IECC 2012 402.3.3.2 Increased SHGC - If checked, will allow for a maximum SHGC of 0.4 for fenestrations that are located no less than 6 feet above the finished floor. This exception only applies to climate zones 1, 2, and 3.

Dimensions - Similar to [Wall Data Input Form](#).

Shading - Shading (external) for windows is activated by checking the shading check box. A set of data input cells become visible where shading parameters can be entered. Shading surfaces include Overhangs, Left and Right Fins. Enter all data that is applicable.

The figure shown below is a basic representation of a window and shading as seen from outside. The dimensions are labeled on the figure and defined below:



Overhang A - Distance from left window edge to left corner of overhang. Units are in feet, 0.0 is the default, and there are no limits.

Overhang B - Distance from top of window to back edge of overhang. Units are in feet, 0.0 is the default, and there are no limits.

Overhang W - Width of overhang. Units are in feet, 0.0 is the default, and the range is 0.0 to no limits.

Overhang D - Depth of overhang. Units are in feet, 0.0 is the default, and the range is 0.0 to no limits.

Overhang Angle - Angle in degrees between overhang and window. When set at 90 degrees, the overhang is perpendicular to the window; if less than 90 degrees, it is tilted downward; if greater than 90 degrees, it is tilted upward. The range is 0 to 180 degrees.

Left/Right Fin A - Distance from left/right window edge to fin back edge. Units are in feet, 0.0 is the default, and there are no limits.

Left/Right Fin B - Distance that top of fin is below top of window. Units are in feet, 0.0 is the default, and there are no limits.

Left/Right Fin H - Height of fin. Units are in feet, 0.0 is the default, and the range is 0.0 to no limits.

Left/Right Fin D - Depth of fin. Units are in feet, 0.0 is the default, and the range is 0.0 to no limits.

[Related Topics](#)



3.6 Door Data Input Form

Door Info

Door Info

Acronym:

Desc:

Select Construct>> Cons1002 - Aluminum door, 1.25 in. polystyrene (U=0.1919 R=5.2104 Den=43.6739 HCapPerSF=0.5344 Abs=0.7000 Emm=0.9 Std Construct)

SetBack Infil Coef Inside Vis Refl

Category

Dimensions

Shape

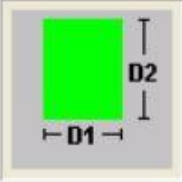
Area: 0

Dimension1:

Dimension2:

Multiplier

Total Area: 0



☒ **Shading**

Shading Data

	Dimension A	Dimension B	Dimension W	Dimension H	Dimension D	Angle
Overhang	0	0	0		0	90
Left Fin	0	0		0	0	
Right Fin	0	0		0	0	

Data inputs are very similar to the [Wall Data Input Form](#).

Data on placement and location, etc, are not required.

Shading - Shading for doors is activated by checking the shading check box. A set of data input cells become visible where shading parameters can be entered. Enter all data that is applicable.

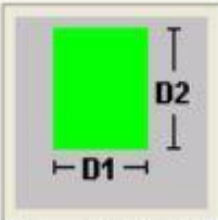
For a detailed account of shading parameters see the [Window Data Input Form](#).

[Related Topics](#)



3.7 Roof Data Input Form

Roof Info

Roof Info	
Acronym	Pr0Zo1Rf1
Description	Pr0Zo1Rf1
Select Construct >>	Cons1047 - Mtl Bldg Roof/R-19 Batt (U=0.0261 R=38.3366 Den=7.7509 HCapPerSF=1.8499 Abs=0.2000 Emm=0.2 Std Construct)
ASHRAE Category	Metal Building Roof
☐	ASHRAE 90.1-2010 Section 5.5.3.1.2 or IECC 2012 402.2.1.1 Exception
Dimensions	
Shape	Rectangle
Dimension1:	100
Dimension2:	100
	
Area: 10000	
Multiplier: 1	
Total Area: 10000	
Skylight Area incl Frame: 100 per Unit; (100 Gross)	
Placement	
Tilt Angle:	0 Horizontal Roof
Location:	Exterior
Sel Adj Zone >>	
Misc	
Gnd Refl	0.2
Inside Vis Refl	0.7
Outside Emiss	0.9
Infl Coef	0.42
Inside Sol Abs	0.3

Data inputs are very similar to the [Wall Data Input Form](#).

ASHRAE 90.1-2010 5.5.3.1.2 or IECC 2012 402.2.1.1 Exception - If checked, will allow the user to bypass minimum solar reflectance and thermal emittance of the roof. This should only be checked if the user is sure that the roof qualifies for an exception.

[Related Topics](#)



3.8 Skylight Data Input Form

Skylight Info

Skylight Info

Acronym:

Desc:

Select Glass Type>>

Desc: ApLbWnd305 --- T11B1C61SK:NC:
Glass type: User Defined Panes=1 SHGC=0.49 U=0.69
VLT=1

☐ Has Curb ☐ Material Plastic (Otherwise assumed Glass) ☐ ASHRAE 90.1-2010 Section 5.5.4.4.2 Exception?

Well Factor Daylighted Area* *See ASHRAE 90.1-2010 Definitions

☐ IECC 2012 402.3.3.4 Increased U-factor ☐ IECC 2012 402.3.3.3 Increased SHGC

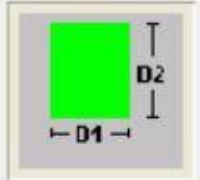
Infl Coef Inside Vis Refl

Dimensions

Shape:

Dimension1:

Dimension2:



Area: 100

Multiplier:

Total Area: 100

Data inputs are similar to [Window Data Input Form](#), except shading is not available for skylights.

Has Curb - If checked, indicates that the skylight is a curb mounted skylight.

Material Plastic (Otherwise assumed Glass) - If checked, indicates that the skylight pane is constructed from plastic instead of glass.

ASHRAE 90.1-2010 5.5.4.4.2 Exception - If checked, indicates that the skylight qualifies for the exception outlines in ASHRAE 90.1-2010 5.5.4.4.2 on maximum SGHC for skylights.

Well Factor - Specifies the total fraction of light entering the light well that penetrates to the room below. This is determined by the geometry of the light well and the reflectance of the surfaces in the light well.

Daylighted Area - Specifies the total floor area that is illuminated by daylight from the skylight.

IECC 2012 402.3.3.4 Increased U-factor - If checked, allows the user to use a larger U-factor for the skylight, provided that the skylight is installed above a daylight zone with automatic daylighting controls.

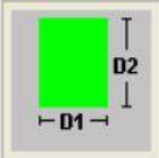
IECC 2012 402.3.3.3 Increased SHGC - If checked, allows the user to use a larger maximum SHGC for the skylight, provided that the skylight is located above daylight zones with automated daylight controls and in the correct climate zone.



[Related Topics](#)

3.9 Floor Data Input Form

Floor Info

Floor Info	
Acronym	Pr0Zo1FI1
Description	Pr0Zo1FI1
Select Construct>	Cons1005 - 1 ft. soil, concrete floor, carpet and rubber pad (U=0.2681 R=3.7300 Den=113.3333 HCapPerSF=34.0000 Abs=0.7000 Emm=0.9 Std Construct)
Dimensions	
Shape Type:	Rectangle
Dimension1:	100
Dimension2:	100
	
Area: 10000	
Multiplier: 1	
Total Area: 10000	
Placement	
Tilt Angle:	180 Horizontal Floor
Location:	Underground
Sel Adj Zone >>	
☐ Use F Value	Underground Floor Type
Enter F Value	0.6
Slab Type	Uninsulated (Default) - 2-ft foundation dep
☐ Heated	☐ Carpeted
Misc	
Gnd Refl	0.2
Inside Vis Refl	0.2
Outside Emiss	0.9
Infl Coef	0.42
Inside Sol Abs	0.8
Inside Film Coeff	0

Data inputs are very similar to [Wall Data Input Form](#)

Some additional inputs associated with underground floor types are:

Underground Floor Type - When the location is chosen as 'Underground', the user should specify the 'Underground Floor Type'.

Slab Type - Specify insulation and geometric characteristics of underground floor type.

Carpeted - Select if the interior floor surface is carpet.

Heated - Select if the slab is heated.

Use F-value - The use F-value check box allows the user to directly enter the perimeter heat loss coefficient for underground floors. In case the F-value of the floor is not known, the user can choose the slab type and default F-values will be assigned to the floor construct by the program.

[Related Topics](#)



3.10 System Data Input Form

Acronym	Description	Variable Air Volume Built-up System--904		No. of Units
Pr0Sy1	System 1			1
*Baseline Cooling Capacity Multiplier		1	*Baseline Heating Capacity Multiplier	1
			*Baseline Fan Capacity Multiplier	1
*Use these multipliers to reduce baseline unmet hours, if any. Ignored for FL code calculations				
Cooling System				Included? ☒
Cooling Capacity [Btu/h]	0			
Reheat Type	NONE			
Source	Purchased Cooling ☐			
Heating System				Included? ☒
Heating Capacity [Btu/h]	0			
Heat Source	Electric Resistance ☒			
	Heating From Plant ☐			
	Purchased Heat ☐			
Air Handling System -Supply				Included? ☒
Supply CFM	1500			
Supply Fan Power [W/cfm]	1.25			
FAN-CONTROL	CYCLING			
MIN-CFM-RATIO	0.3			
Allowable Name Plate				

Compliance Feedback set to: 2009 INTERNATIONAL ENERGY CONSERVATION CODE

Acronym - A short description (up to 20 characters)

Description - A detailed description (up to 50 characters)

No. of Units - Enter the number of such units (Use this to input multiple systems serving one or more Zones)

Baseline Cooling Capacity Multiplier, Baseline Heating Capacity Multiplier and Baseline Fan Capacity Multiplier - If the unmet hours for the **baseline (or reference building)** needs adjusting to meet the unmet hour rule, (See ASHRAE 90.1 or 2014 Florida Code for details) these three parameters serve as capacity multipliers for the baseline building to adjust system output. These parameters will not effect the unmet hours for the proposed building, these hours can be met by modifying the capacity of the proposed system itself.

Component Name - The type of the component (whether cooling, heating etc) is displayed. If a check box appears near the name of the component, it indicates that the component is optional and user decided whether to select it or not.

Cooling System		Included?
Cooling Capacity [Btu/h]	600000	
Cooling EER or SEER	15	
Cooling Efficiency Type	EER	
Integrated Part Load Value	11.2	
Reheat Type	None	
Condensor Type	Air Cooled ☒	
	Water Cooled ☐	
	Evaporatively Cooled ☐	

Input Parameter - Shows the different required parameters (like capacity, EER, IPLV etc.) for the respective component.

Cooling Capacity [Btu/h]	600000	Parameter
Cooling EER or SEER	15	
Cooling Efficiency Type	EER	
Integrated Part Load Value	11.2	
Reheat Type	None	

Options - Describes the option type for the component. Any sub-option may be selected by clicking on the sub option.;

Option

Condensor Type	Air Cooled	☒
	Water Cooled	☐
	Evaporatively Cooled	☐

Sub-Option Choice

Sub-Option Parameter - If parameter field appears next to a selected sub option, the value of the parameter must be selected.

Sub-Option Parameter	Sub-Option Parameter Value
Heating Efficiency, Et/Ec, AFUE	95
Heating Efficiency, Et/Ec, AFUE	78
Heating COP	3.1

Steps for Entering Data

To select an optional component click on the check box to select it.

Click on the corresponding column. A pop up window will appear (for example: the figure below displays the selection of fan power option). Enter or select the value and then click OK.

To choose a particular option in a component click on the radio button.

Form1

Allowable Name Plate HP

Fan Brake Horse Power

OK

Cancel

Description

Constant Volume Packaged System

System 2

No. of Units

1

Furnace - Oil		Heating Efficiency, Et/Ec, AFUE	78
Heat Pump - Electric		Heating COP	3.1
Heating From Plant			

Air Handling System - Supply

Included?

Supply CFM	9000		
Fan Power (W/cfm)	0.5		
Fan Power Option	Allowable Name Plate HP		
Fan Power Allowance	0		

Air Handling System - Return

Included?

Return CFM	1500		
Return Fan Power (W/cfm)	0.8		
Fan Power Option	Allowable Name Plate HP		
Fan Power Allowance	0		

Pr0Zo1Wa4

Windows

Pr0Zo1Wa4

Doors

Roofs

Pr0Zo1Rf1

Skylights

Pr0Zo1Rf1S

Floors

Pr0Zo1Ff1

Systems

Pr0Sy2

Pr0Sy4

Acronym	Description	Constant Volume Packaged System	
PrOSy2	System 2	No. of Units: 1	
Return CFM	1500		
Return Fan Power (W/cfm)	0.8		
Fan Power Option	Allowable Name Plate HP		
Fan Power Allowance	0		
Air Distribution System (Sup)		Included? ☒	
Air Distribution Location	Conditioned		
ADS Heating/Cooling Type	Combined Heating & Cooling Ducts		
ADS Rvalue	6		
Air Distribution System (Ret)		Included? ☐	
Air Distribution Location	Conditioned		
ADS Heating/Cooling Type	Combined Heating & Cooling Ducts		
ADS Rvalue (Ret)	0		
Energy Recovery Unit		Included? ☐	

To see compliance feedback, go to Menu: Tools > Settings > Feedback Settings and set Feedback Option

Air Distribution System - If checked, compliance for this component will be checked. Required inputs are "**Location**" and "**R-Value**".

Energy Recovery Unit - If checked, ERV credit will be applied. Applicable only for the 2010 Florida Code (see the 2010 Florida code for details of credit).

[Related Topics](#) 

3.11 Plant Data Input Form

Plant Data for Chillers & Boilers

EnergyGaugeSummit Help

	Type	Use	Equipment	Capacity	Qty	Efficiency	IPLV	Compliance	Non-Standard?	Use Path B?
1	Cooling Equipment	☒	Open centrifugal chiller	20.00 [Tons]	1	5.000 [COP]	5.250 [COP]	-		
2	Cooling Equipment	☐	Open reciprocating chiller					-		
3	Cooling Equipment	☐	Hermetic Centrifugal chiller					-		
4	Cooling Equipment	☐	Hermetic screw or scroll chiller					-		
5	Cooling Equipment	☐	Thermal Storage					-		
6	Cooling Equipment	☐	Cooling Tower					-		
7	Heating Equipment	☐	Steam Boiler (Fuel)					-		
8	Heating Equipment	☐	Hot Water Boiler (Fuel)					-		
9	Heating Equipment	☐	Electric Steam Boiler					-		
10	Heating Equipment	☐	Electric Hot Water Boiler					-		
11	Heating Equipment	☐	Furnace					-		
12	Water Heating Equipment	☐	Domestic hot-water heater					-		
13	Water Heating Equipment	☐	Electric domestic hot-water heater					-		

	Parameter	Value
1	Chilled water pump type, fixed or variable speed	Fixed Speed
2	Open Chiller Leaving Evap Temp in [F]	44 [F]
3	Open Centrifugal Chiller Flow/ton	3 gpm/ton
4	Open Chiller LIFT in [F]	25 [F]
5	Condenser type, open centrifugal chiller	Air
6	Fan electric power to centrifugal chiller capacity ratio	0.03

A plant is added or removed from the project by checking or unchecking the appropriate 'Use' check box. Data that must be entered appears. Based on the plant selected, additional parameters that must be input appears in the parameter grid below.

Type - Shows the type of equipment.

Equipment - Displays the name of the equipment.

Capacity - Specifies the capacity of the selected equipment. The units in which the data is to be entered is also shown.

Quantity - Specifies how many units are installed.

Efficiency - Specifies the efficiency of the equipment. The units in which the data is to be entered is also shown.

IPLV - Specifies Integrated Part Load Value, if required.

Non-Standard? - Applies only to Open Centrifugal Chillers and Hermetic Centrifugal chillers. If checked, the corresponding plant equipment is considered non-standard and the efficiency requirement for non-standard chillers apply (see ASHRAE 90.1 or 2010 Florida code for details). Non-standard chillers require additional parameters to be input such as Flow/ton in gpm; LIFT in o F;

Note for Cooling Tower: The cooling tower requires the following input parameters: Performance gpm/hp; Fan type[Propeller or Axial or Centrifugal]; Tower Type[Open or Closed circuit]

Plant Data for Water Side Systems Only

Plant Data for Chiller/Boilers				Plant Data for Water Side Systems only			
Type	Use	Equipment	Qty	Compliance	Non-Standard?	Use Path B?	
1 Cooling Equipment	☒	Cooling Tower for Water side system (WSS)	1	Failed (MinEff=38.2 (gpm/hp))			
2 Heating Equipment	☒	Boiler for Water Loop Heat Pump	1	Passed (MinEff=80 AFUE)			

Parameter	Value
1 Cooling Tower Performance (gpm/hp)	7 gpm/hp
2 Cooling Tower Fan Type	Propeller Fan
3 Cooling Tower Type	Open Circuit
4 Circulating Pump Head	60 ft
5 Circulating Pump Motor Efficiency	0.9
6 Circulating Pump Type	Fixed Flow
7 Circulating Pump Impeller Efficiency	0.77
8 Tower Impeller Efficiency	0.77
9 Cooling Tower Size	0 Ton
10 Tower Number of Cells	0
11 Tower Cap Control	Single Speed Fan
12 Tower Pump Pressure Head	20 ft
13 Tower Motor Efficiency	0.9
14 Tower Design Approach Temperature	7 F
15 Boiler Capacity (Millions of Btu)	0 Mill Btu
16 Boiler Fuel	Natural Gas Boiler
17 HP Boiler Efficiency	80 AFUE/Ec/Et

These are water-side equipment. Use these equipment for Waterloop Heat pump systems and packaged systems with watercooled condensers.

For built up systems (eg. non packaged) served by central plant, use regular plant equipment in the first tab

Compliance Feedback: set to: IECC 2009
Note: Inputs for Include in Simulation and Draw-Gallons per Minute fields may be ignored for Florida code compliance calculation

A plant servicing a water-side system is added or removed from the project by checking or unchecking the appropriate 'Use' check box. Data that must be entered appears. Based on the plant selected, additional parameters that must be input appears in the parameter grid below. Please note that these entries are specifically meant for heating/cooling systems that have a water-side component; such as a packaged system with water source cooling or a heating system with water loop heating.

Type - Shows the type of equipment.

Equipment - Displays the name of the equipment.

Quantity - Specifies how many units are installed.

Compliance - If feedback controls are enabled, this will display if the current efficiencies are passing the selected code efficiency minimums.

Non-Standard? - Applies only to Open Centrifugal Chillers and Hermetic Centrifugal chillers. If checked, the corresponding plant equipment is considered non-standard and the efficiency requirement for non-standard chillers apply (see ASHRAE 90.1 or 2010 Florida code for details). Non-standard chillers require additional parameters to be input such as Flow/ton in gpm; LIFT in o F;



[Related Topics](#)

3.12 Water Heater Data Input Form

Category	Desc	Capacity	Installed Number	Input Rate	Draw Pattern	Efficiency	Loss	Compliance	Include in Simulation	Draw-Gallons per Minute
1	Electric Storage water heater	WH#1 50.00 [Gal]	1	15.00[kW]	Any		0.3000 %	Passed (MaxLoss=84.)	☐	0.0000
2	Gas Storage water heater	WH#2 50.00 [Gal]	1	80000.00[Btu/h]	Any	0.8800 [Ei/Et]	250.0000 Btu/h	Passed (MinEff=8,MaxLoss=877.82.)	☐	0.0000

Any
Very Small
Low
Medium

Water Heater Category - This field describes the various Water Heater categories available. They are available in the drop down box. Select one from the [list](#).

Description - You can type your own description here.

Capacity - The capacity of the water heater. Units are displayed in the cell.

Input Rate - The power rating of the water heater in the appropriate units displayed.

Draw Pattern - Allows you to choose the draw pattern used by the Water Heater. For 2020 code choose the draw pattern for older code calculations use the 'Any' option.

Efficiency - The efficiency of the water heater between 0 and 1, entered as a decimal (between 0 and 1).

Loss - The loss in the appropriate unit displayed.

Compliance - Shows whether water heater complies with code or not.

Note: Not all inputs may be required for all water heater types.

See the [Section on Working with Grids](#) to add, delete and modify data.

Related Topics



3.13 External Lighting Data Input Form

External Lighting Zone

All other areas

All other areas

	Category	Desc	Number of Luminaires	Watts per Luminaire	Lumin Eff [lm/Watt]	Area or Length	Control Type
1	Parking Lots and Drives (Uncovered)	Ext Light 1	1	40.00 Watts	60.00	1000.00 SF of Area	Astronomical Timer Control
2	Exit Signs	Ext Light 2	1	35.00 Watts	60.00	8.00 Total No. of faces	Other Independent Control

Compliance Feedback set to: No Feedback

Compliance check includes a excess/Base allowance of 9.50(w)

Compliance: Passed --- Total: 75 (w) Tradable Budget: 40 (w) Tradable Allowance: 159.5 (w)

External Light Category - External lighting category. Select one from the [list](#).

Description - You can type your own description or comment here.

No. of Luminaires - Specify the number of light fittings in that space.

Watts per Luminaire - Specify the lighting wattage per fitting.

Luminaire Efficiency - Specify the efficiency of each luminaire.

Area or Length - Input depends on the category selected. Enter length or area.
Required units will be shown.

Control Type - Specify the type of control for the luminaire from a drop down list.

See the section on [Working with Grids](#) to add, delete and modify data.

[Related Topics](#)



3.14 Piping Data Input Form

	Type	Operating Temperature	Insulation Conductivity [Btu-in/(h.ft ²)]	Nominal Pipe Diameter [in]	Insulation Thickness [in]	Check if Runout up to 4 ft	Compliance
1	Heat Sys Piping	105.00	0.28	0.25	0.60	☐	Passed (Calc M)

Compliance Feedback set to: No Feedback

Type - Specify the Piping application.

Operating Temperature - The operating temperature at which the pipe will operate.

Insulation Conductivity - Specify the conductivity of the insulation around the pipe.

Nominal Pipe Diameter - Specifies the diameter of the pipe.

Insulation Thickness - Specify the thickness of the insulation around the pipe.

Check if Runout up to 4 ft - Specific if the pipe is a runout.

Note: Units are shown on the form.

See the section on [Working with Grids](#) to add, delete and modify data

[Related Topics](#)

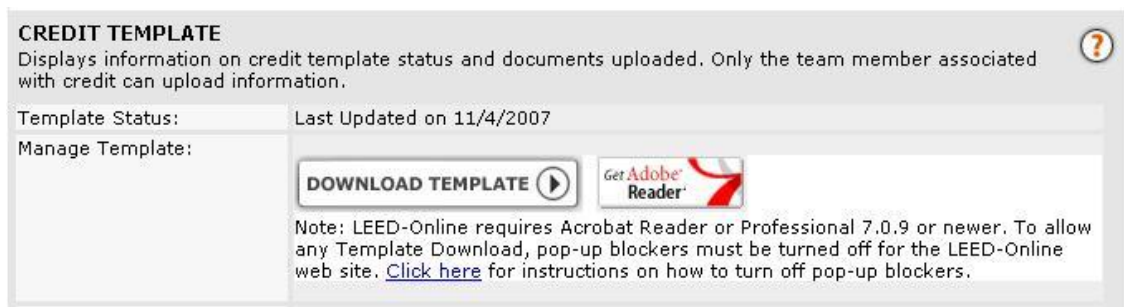


3.15 LEED22 Additional Inputs Form

The LEED NC22 additional inputs form is only required to be filled up by users planning to run the LEED 2.2 New Construction (NC) calculation on the proposed (design) building. The results from the LEED 2.2 NC simulation can be submitted, via an existing PDF template, to the US Green Building Council (USGBC) website for documentation of the EA Credit 1 (EA C1) USGBC points that can be claimed by users when completing the LEED certification process. The required empty PDF template needs to be downloaded by the user from the USGBC website prior to running the LEED 2.2 NC simulation. The path for the location of this empty PDF template is to be given in the form as shown in the figure below.

Once this additional information has been entered by the user for a particular project/building, it will be stored as part of that EGC file. The remaining information for filling the USGBC simulation data PDF template will be automatically generated by the EnergyGauge Summit software and pulled into the PDF template every time the LEED simulation is run. The PDF template should be downloaded from the USGBC website as shown in the figure.

USGBC website PDF Template Download:



The EnergyGauge form is explained below.

General Info Tab:

The screenshot shows the 'General Info' tab of the EnergyGaugeSummit application. The form is divided into three sections: 'General Info', 'On Site Renewable', and 'Exceptional Calculation'. The 'General Info' section contains the following fields and controls:

- First Name:** John
- Last Name:** Doe
- Date:** 12/29/2008 (dropdown menu)
- User Name:** john_doe@doe.com
- Password:** XXXXXXXXXXXXXXXX (masked)
- Company:** John Doe & Associates LLC
- Seeking Alternate Compliance Points:** ☐ (checkbox)
- Alternate Compliance Points:** 0 (spin button)
- Target Score Finder:** 0 (spin button)
- Alternate Compliance Points Explanation/Document(s):** (empty text area)
- Supporting Documentation listed below:** ☒ (checkbox)
- SupportingDocs.pdf:** (text area containing the filename)
- Select EAC1 Template File:** C:\D-Drive\EGC2004a\Samples\TemplateRenderer.pdf (text field)

The menu item 'Edit' > 'LEED 22NC Additional Inputs' brings up the LEED 22 additional information form. The first tab to be filled out is the General Information tab. This requires the user, wanting to run the LEED 22 NC simulation and submit their EA Credit 1 data to the online USGBC web service, to enter the information like their user name, password and so on. This information is required to populate the PDF file that is submitted to claim USGBC credit points for energy optimization of the proposed building.

On Site Renewable Tab:

General Info		On Site Renewable		Exceptional Calculation	
☐ Includes On Site Renewable Energy		On site Renewable Calculation Method ☒ Auto Calc ☐ Separate Analysis ☐ Direct Modelling			
Item	Renewable Source	Annual Energy Generated [kW/h]	Rated Capacity	Renewable Energy Cost	Backup Energy Type

The on-site renewable energy generation and cost information is also required to calculate USGBC credit points and this information is provided by the user in this form. The information provided here is automatically inserted in the PDF file that should be submitted to the USGBC web service. Clicking the '**Include On Site Renewable Energy**' check-box will allow the user to enter details about the renewable energy sources in the proposed building.

Exceptional Calculation Tab:

General Info		On Site Renewable		Exceptional Calculation	
☐ Includes Exceptional Calculation Methods		Exceptional Calculation Method ☒ Automatic Calculation ☐ Separate Analysis			
Exceptional Calculation Measures					
	Item	Short Description	Narrative		
Exceptional Calculation Measure Details					

This tab is used to document any exceptional calculation methods that may have been used by the user as defined in the ASHRAE 90.1 2004 Appendix G. The information provided here is automatically inserted in the PDF file that should be submitted to the USGBC web service. Clicking the '**Include Exceptional Calculation Methods**' checkbox will allow the user to enter details about the exceptional calculation methods used in modeling the proposed building as allowed in ASHRAE 90.1 2004 Appendix G.

[Related Topics](#)



3.16 LEED Online Submission Instructions

Please take a minute to review these instructions before running and submitting a LEED calculation using EnergyGauge Summit v 3.22.

Pre-submittal tasks

EnergyGauge is about to enter all your data in your LEED EA Credit 1 template. Please review all the information on the template before submission.

NOTE: Do not edit the PDF template displayed by the software as that data is not saved. Display is for review purposes only

If you need to make any changes to the inputs, please close the template and make the changes in the EnergyGauge inputs screens

Post-submittal tasks

Once you have submitted your LEED EA Credit 1 data through the EnergyGauge software, please download your EAc1 PDF template for that project from the LEEDOnline website. If your submission was successful, this template should now contain all the outputs from the LEED simulation runs.

Review the template for any incorrect entries and enter all additional entries not handled by EnergyGauge. These include, but are not limited to:

- **External Lighting Data** - Annual external lighting energy use must be added for both proposed and baseline buildings as new entries in tables 1.8.1 and 1.8.2. This value should generally be the same for both the proposed and baseline buildings unless otherwise justified.
- **Water Heating Data** - EnergyGauge does not currently simulate annual energy use for certain water heaters. If an entry for domestic hot water use is missing from table 1.8.1 and 1.8.2, then such an entry must be added at this point.
- Any other missing data that may be required by the USGBC should be added into the template at this point.

The PDF template should now be re-submitted manually (not through EnergyGauge) for final review of the project by the USGBC.

Any changes/additions made to the template after its submission through the EnergyGauge software will have to be repeated should you require to re-run the calculations using the EnergyGauge software.



3.17 District Thermal Energy Management for LEED

EnergyGauge Summit allows users to perform the steps required to incorporate district heating and cooling in their proposed building as per requirements from USGBC. Since EnergyGauge will automatically create the ASHRAE Appendix G baseline building against which the proposed building is evaluated to calculate LEED points, the baseline building requirements when the proposed building has district cooling and heating are also handled automatically by the software.

Calculating LEED district heating and cooling is a two step process as outlined in the documentation provided on the USGBC website.

District Thermal Energy Management - STEP 1:

If the user's proposed building contains district heat and cooling, the first step requires the user to create a proposed building in EnergyGauge using any of the 3 available systems that allow purchased heating and cooling to be specified. The three systems are:

- Two pipe fan coil system (system ID 201)
- Four pipe fan coil system (system ID 202)
- Variable volume built-up system (system ID 904)

The user must also add the purchased heating and cooling cost in the utilities and climate tab as shown in figure below.

All these data not required for Florida Code Compliance

Select by Location Data

Latitude Longitude

Cooling Degree Days Heating Degree Days

Summer Design Temp Winter Design Temp

Altitude

Choose County, if listed here: Weather Date Set:

Climate Location: ALACHUA COUNTY, ALACHUA COUNTY, FL
 Weather file: C:\Program Files (x86)\Common Files\EnergyGauge\Weather\USA\TMY3\FL_GAINESVILLE_REGIONAL_AP.tm3

Utility Rates ☐ Use these Rates for 2010 Florida Code Instead of Defaults

	Fuel	Fuel Type	Description	Monthly Charge	Energy Charge (\$)	Energy Charge Unit	Demand Charge (\$/Unit)	TOU Schedule
1	1	Electricity	Res1	\$ 0.0000	\$ 0.1000	kWh	\$ 0.0000	None
2	2	Natural Gas	Res2	\$ 0.0000	\$ 0.5000	therm		None
3	3	Fuel oil	Res3	\$ 0.0000	\$ 1.0000	gal		None

Once the utilities have been included, the user needs to specify the proposed building features including all the HVAC systems, building envelope components, internal equipment and other proposed building inputs including user entered profiles and

schedules as applicable to the proposed building. When all other additional input LEED requirements have been entered, the user should run the **‘LEED EAc1 Online Submittal’** menu from the **‘Calculate’** menu option. This will run the LEED EAc1 simulation for the proposed building. The software will automatically create the baseline building for Step 1 of the LEED simulation. The baseline building will also use the same systems and purchased heating and cooling utility rates as specified for the proposed building.

The user should also include in the proposed model of the building an on-site local plant that will act as a virtual substitute for the district thermal heating and cooling plant. The efficiencies of this on-site virtual plant should match that of the district heating and cooling set-up as per official USGBC requirements.

District Thermal Energy Management - STEP 2:

USGBC requires that the Step 2 of simulation be run only if the user’s proposed building model is able to meet and exceed a minimum of 2 EAc1 LEED points as compared to the Appendix G baseline building energy use. If this requirement is satisfied when the first step of the simulation is run, the LEED EAc1 PDF output for the can be saved by the user as shown in figure below.

Please fill out the following form. Highlight Fields

LEED-NC
LEED FOR NEW CONSTRUCTION

LEED-NC 2.2 Submittal Template
EA Credit 1: Optimize Energy Performance

NARRATIVE (Optional)
Please provide any additional comments or notes regarding special circumstances or considerations regarding the project's credit approach.

☐ The project is seeking point(s) for this credit using an alternate compliance approach. The compliance approach, including references to any applicable Credit Interpretation Rulings is fully documented in the narrative above. (Indicate the number of points documented in the "Alternative Compliance Points Documented" field below).

0 Alternative Compliance Points Documented

Project Name: Physical Science

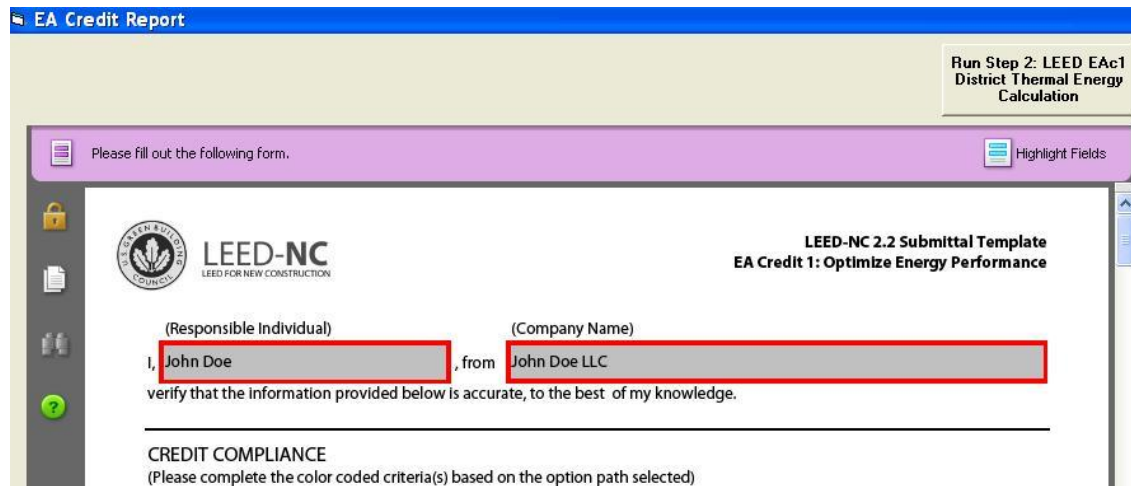
Credit: EA Credit 1: Optimize Energy Performance Points Documented: 10

READY TO SAVE THIS TEMPLATE TO LEED-ONLINE? Please enter your first name, last name and today's date below, followed by your LEED-Online Username and Password associated with the Project listed above to confirm submission of this template.

John	Doe	11/3/2007	gngdntehr	*****
First Name	Last Name	Date	Username (Email Address)	Password

SAVE TEMPLATE

The saved output from Step 1 is required as supporting documentation for the Step 2 results when the final submission for the LEED points is done. Once the Step 1 output is saved, a button to run Step 2 of the LEED simulation will become visible as shown in the figure below. Running Step 2 will then create a new proposed building model with modified HVAC systems running of the local virtual plant that is entered before Step 1 is initiated. The baseline building model for Step 2 is calculated as per ASHRAE Appendix G requirements.



EA Credit Report

Run Step 2: LEED EA1 District Thermal Energy Calculation

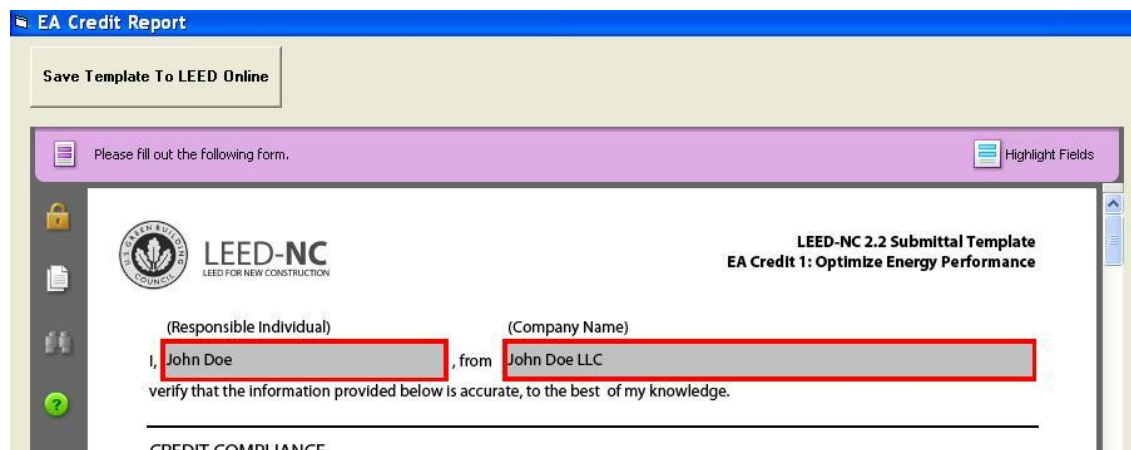
Please fill out the following form. Highlight Fields

LEED-NC 2.2 Submittal Template
EA Credit 1: Optimize Energy Performance

(Responsible Individual) (Company Name)
I, John Doe, from John Doe LLC
verify that the information provided below is accurate, to the best of my knowledge.

CREDIT COMPLIANCE
(Please complete the color coded criteria(s) based on the option path selected)

The EnergyGauge software will then display the LEED EA1 PDF output for Step 2 as shown in figure below. The output can directly be uploaded to the LEEDOnline website by clicking the 'Save template to LEEDOnline' button on the interface.



EA Credit Report

Save Template To LEED Online

Please fill out the following form. Highlight Fields

LEED-NC 2.2 Submittal Template
EA Credit 1: Optimize Energy Performance

(Responsible Individual) (Company Name)
I, John Doe, from John Doe LLC
verify that the information provided below is accurate, to the best of my knowledge.

CREDIT COMPLIANCE

The user is required to submit all other supporting documentation to the LEEDOnline website as per USGBC requirements.

[Related Topics](#)



3.18 Profiles and Schedules

EnergyGauge Summit allows user created profiles and schedules for all non-code type simulations. For code calculations, default schedules and profiles, as required by the particular standard will be used.

Profiles are assigned directly to zones or the building and they consist of individual annual schedules for all the following inputs:

- People
- Lighting
- Infiltration
- Equipment
- Sources
- Heating Set-point
- Cooling Set-point
- Hot Water schedule
- Heating Schedule
- Cooling Schedule
- Fan operation Schedule

There is a Master library for profiles and schedules and a Project library for each individual project. The Master library can be accessed from the '**Resources**' whereas the Project library can be accessed from the '**Edit**' menu option. A user can add a profile or a schedule from the Master library to the Project library of a new profile or schedule can be specified by the user in the Project library. The Master and Project libraries are shown in figures 1 and 2 below, respectively.

EnergyGaugeSummit Help

Project Profiles and Schedules

Profiles Schedules Hourly Schedules

	Id	Acronym	Description	Heating Design Temp	Cooling Design Temp
1	0	No Classification	No Classification	72.00	72.00
2	2	Office (Business)	Office (Business)	72.00	72.00
3	5	School (educational)	School (educational)	72.00	72.00
4	6	Retail	Retail	72.00	72.00
5	7	AssemblyFL	FlaCom Assembly	72.00	72.00

	Id	Category	Schedule
1	201	People	Fractional Null Schedule (2)
2	202	Lighting	Fractional Null Schedule (2)
3	203	Infiltration	Fractional Null Schedule (2)
4	204	Equipment	Fractional Null Schedule (2)
5	205	Sources	Fractional Null Schedule (2)
6	206	HeatTemp	Set Point 55 (202)
7	207	CoolTemp	Set Point 99 (201)
8	208	Hot Water Schedule	Fractional Null Schedule (2)

Add Profile 0 to Current Project

Profiles Schedules

	Id	Acronym	Description	Heating Design Temp	Cooling Design Temp
1	0	No Classification	No Classification	72.00	78.00
2	501	ACM-NonRes	ACM Nonres	72.00	78.00
3	1001	School (educational)	School (educational)	72.00	72.00

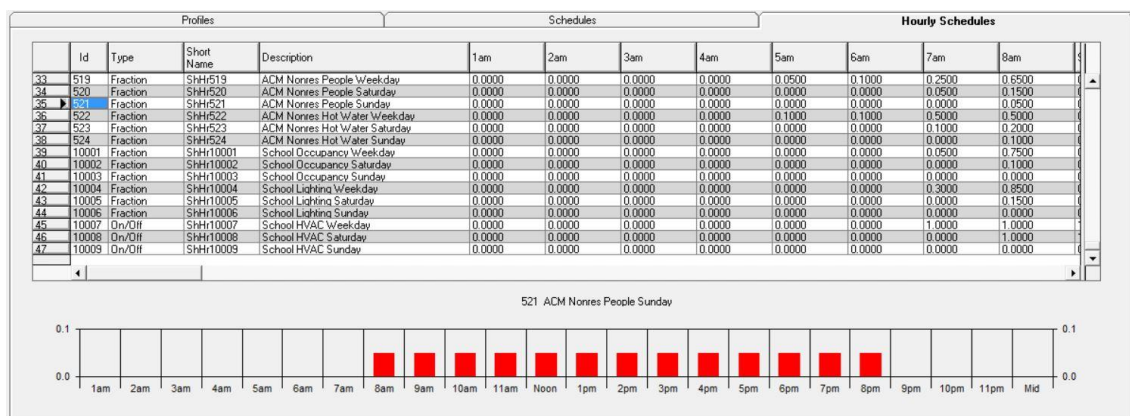
	Id	Category	Schedule
1	201	People	Fractional Null Schedule (2)
2	202	Lighting	Fractional Null Schedule (2)
3	203	Infiltration	Fractional Null Schedule (2)
4	204	Equipment	Fractional Null Schedule (2)
5	205	Sources	Fractional Null Schedule (2)
6	206	HeatTemp	Set Point 55 (202)
7	207	CoolTemp	Set Point 99 (201)
8	208	Hot Water Schedule	Fractional Null Schedule (2)

Right clicking in the first input box allows the addition of a new profile to the Project library. If this profile is from the Master library, all the default annual schedules for that profile for the sub-categories mentioned above will be added to the Project library. The user can then assign this profile to a particular zone or to the entire building. The user can also choose to make changes to this profile by replacing or modifying the annual schedules from the selected profile.

Each annual schedule for people or lighting or equipment or other sub-categories mentioned above, consists of periods for a single year. The user can have as many periods in the year as required, so long as there is no duplication or overlapping and the starting date of one period is January 1 and the end date is December 31. Figure 3 below shows the addition of periods to an annual schedule.

[illegible]

Hourly Schedules:





4. Master Library Information

4.1 Master Materials Library Form

The Master Materials Library is a list of over 300 materials used in building construction. This serves as a resource from which selected materials can be inserted into your project.

The user is not allowed to make any changes.

	Id	Desc	Thick [in]	Conductivity [Btu/in.F]	Density [lb/cf]	Sp. Heat [Btu/lb.F]	Use	Category	Thickness Fixed	Use RVal	R-Val [h.ft.F/Btu]
1	1	Outside surface resistance	0.000	0.000	0.00	0.00	Exterior Finish	Air Layer	☒	☒	0.330
2	2	1 in. Stucco	0.083	0.400	16.00	0.20	Exterior Finish	Stucco	☒	☒	0.208
3	3	4 in. Face brick	0.333	0.770	125.00	0.22	Exterior Finish	Brick	☒	☒	0.432
4	4	Steel siding	0.005	26.000	480.00	0.10	Exterior Finish	Aluminum or Ste	☒	☒	0.000
5	5	1/2 in. Slab	0.042	0.110	70.00	0.40	Exterior Finish	Stucco	☒	☒	0.379
6	6	Outside surface resistance	0.000	0.000	0.00	0.00	Exterior Finish	Air Layer	☒	☒	0.330
7	7	Gyp or Plaster Finish	0.042	0.240	78.00	0.26	Interior Finish	Gypsum or plast	☒	☒	0.174
8	8	4 in. Face brick	0.333	0.770	125.00	0.22	Exterior Finish	Brick	☒	☒	0.432
9	9	Air space resistance	0.000	0.000	0.00	0.00	Insulation	Air Layer	☒	☒	0.910
10	10	1 in. Insulation	0.083	0.025	2.00	0.20	Insulation	Insulation	☒	☒	3.320
11	11	2 in. Insulation	0.167	0.025	2.00	0.20	Insulation	Insulation	☒	☒	6.680
12	12	3 in. Insulation	0.250	0.025	2.00	0.20	Insulation	Insulation	☒	☒	10.000
13	13	1 in. Insulation	0.083	0.025	5.70	0.20	Insulation	Insulation	☒	☒	3.332
14	14	2 in. Insulation	0.167	0.025	5.70	0.20	Insulation	Insulation	☒	☒	6.680
15	15	1 in. Wood	0.083	0.070	37.00	0.39	Structural	Wood	☒	☒	1.190
16	16	2 in. Wood	0.167	0.070	37.00	0.39	Structural	Wood	☒	☒	2.386
17	17	4 in. Wood	0.333	0.070	37.00	0.39	Structural	Wood	☒	☒	4.752
18	18	2 in. Wood	0.167	0.070	37.00	0.39	Structural	Wood	☒	☒	2.386
19	19	3 in. Wood	0.250	0.070	37.00	0.39	Structural	Wood	☒	☒	3.571

See the section on Working with Grids to add, delete, and modify data.

Viewing Selected Materials

You can view selected materials by using the short-list radio buttons provided on the form.

Show All - Clicking this radio button will display all materials in the library.

Show by Material Use - Clicking this radio button will allow materials to be short-listed by general use of the material. After clicking this radio button, you must select the appropriate use category from the Combo box below the radio button.

Show by Material Category - Clicking this radio button will allow materials to be short listed by general type of materials. After clicking this radio button, you must also select the appropriate type from the Combo box below the radio button.

Fields

Description - Is the description about the material

Thick Specifies the thickness of the material in feet

Conductivity - Specifies the thermal conductivity of the material

Density - Specifies the material density

Specific Heat - Specifies the specific heat of the material

Use - Specifies the general use for the material

Category - Specifies material type

Fixed Thickness - Not Currently Implemented

Use R-Val - Two possible specifications for a material are possible:

- 1) Specify valid thickness and all thermal properties. In this case, valid conductivity, density and specific heat must be specified and the 'Use RVal' check box must be unchecked. This option is used for materials that are likely to have mass effects or when these properties are accurately known. The program will use the thermal properties and ignore the R-Value.
- 2) For light and thin materials, or for those materials for which the thermal properties are not available, the second option is to specify only the R-Value. In this case the 'Use R-Value' box must be checked. The program will ignore the thermal properties, and use only the R-Value and treat the material as having no mass.

R-Value - Specifies the thermal resistance of the material

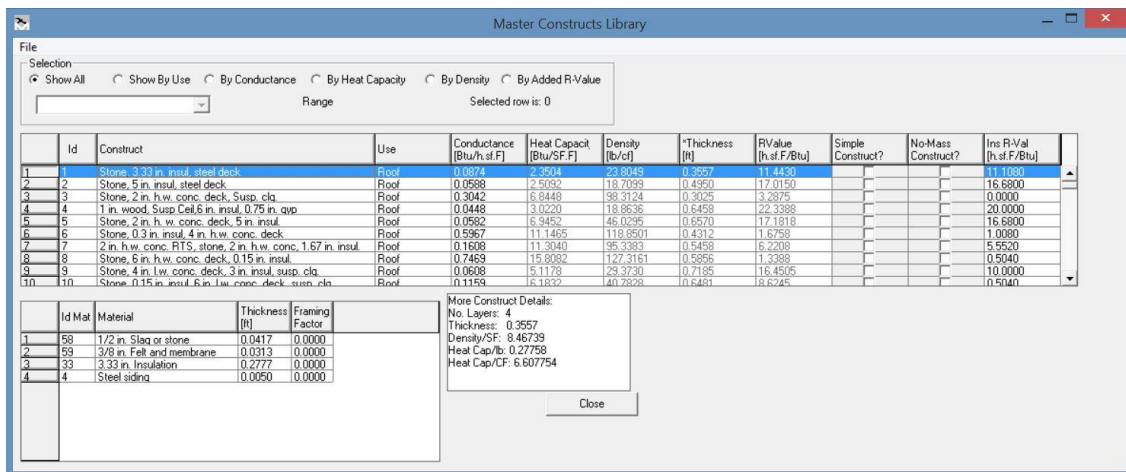
The user is not allowed to make any changes.



4.2 Master Constructs Library Form

The Master Constructs Library is a list of over 300 constructs used in building construction. This serves as a resource from which selected constructs can be inserted into your project.

The user is not allowed to make any changes.



See the section on [Working with Grids](#) to add, delete, and modify data.

Viewing Selected Materials

You can view selected materials by using the short-list radio buttons provided on the form.

Show All - Clicking this radio button will display all constructs in the library.

Show by Use - Clicking this radio button will allow constructs to be short-listed by general use of the material. After clicking this radio button, you must select the appropriate use category from the Combo box below the radio button.

By Conductance - Clicking this radio button will allow short listing by a range of conductance. After clicking this radio button, you must also select the appropriate range from the Combo box below the radio button.

By Heat Capacity - Clicking this radio button will allow short listing by a range of heat capacity. After clicking this radio button, you must also select the appropriate range from the Combo box below the radio button.

By Density - Clicking this radio button will allow short listing by a range of density. After clicking this radio button, you must also select the appropriate range from the Combo box below the radio button.

By Added R-Value - Clicking this radio button will allow short listing by a range of R-Value. After clicking this radio button, you must also select the appropriate range from the Combo box below the radio button.

Fields

Use - Specifies the possible use for the construct

Conductance - Specifies the thermal conductivity of the construct

Heat Capacity - Specifies the heat capacity of the construct

Density - Specifies the construct density

Thickness - Specifies the total thickness of the construct

R-Value - Specifies the thermal resistance of the construct

Simple Construct - See explanation under [Project Materials and Constructs](#)

No-Mass Construct - See explanation under [Project Materials and Constructs](#)

Constructs Library Layers Table

The subtable provides the details of the layers in the selected construct. The layers are specified starting from the outside layer to the inside layer. That is the outermost layer is the first layer.

ID - Specifies the ID of the material the layer is made of. It corresponds to a valid material number from the [Master Material Library](#).

Material - Specifies the name of the material the layer is made of. It corresponds to a valid material name from the [Master Material Library](#).

Thickness - Specifies the thickness of the layer. When a layer is added, the thickness of the material, as specified in the material library, is used. However, an overriding thickness can be entered here, in which case the original thickness in the material library is ignored and the value entered is used.

Framing Factor - Specifies the graming, if any, of the layer. Framing Factor is the ratio of the framing area on the layer to the total area of the layer.

The user is not allowed to make any changes.



5. User Library Information

5.1 Project Materials and Constructs

Library Form

Constructs Tab

The Project Constructs Library is very similar to the Master Constructs Library except that it is associated with a specific Project that is currently open or created. This library stays with the Project and changes made here are applicable only to the current project.

The user may make any modification as required.

Constructs			Materials					
	Mat No	Desc	Conductivity [Btu/h.ft.F]	Density [lb/cf]	SpecificHeat [Btu/lb.F]	Thick [ft]	R-Value [h.ft²/Btu]	Use R-Value
1	264	ALUMINUM, 1/16 IN	26.0000	480.0000	0.1000	0.0050	0.0002	
2	214	POLYSTYRENE, EXP., 1-1/4 IN.	0.0200	1.8000	0.2900	0.1042	5.2100	
3	187	GYP OR PLAS BOARD, 1/2 IN	0.0920	50.0000	0.2000	0.0417	0.4533	
4	178	CARPET W/RUBBER PAD	0.0000	0.0000	0.0000	0.0000	1.2300	✓
5	285	Sol. 1 ft.	0.5000	100.0000	0.2000	1.0000	2.0000	
6	48	6 in. Heavyweight concrete	1.0000	140.0000	0.2000	0.5000	0.5000	
7	267	0.75" stucco	0.4000	16.0000	0.2000	0.0625	0.1563	
8	215	POLYSTYRENE, EXP., 2 IN.	0.0200	1.8000	0.2900	0.1867	8.3250	
9	105	CONC BLK HW, 8 IN, HOLLOW	0.0880	68.0000	0.2000	0.8867	1.1002	
10	256	WOOD, SOFT, 1-1/2 IN	0.0660	32.0000	0.3300	0.1250	1.8539	
11	86	BRICK, COMMON, 4 IN	0.4160	120.0000	0.2000	0.3333	0.8012	
12	23	6 in. Insulation	0.0250	5.7000	0.2000	0.5000	20.0000	
13	4	Steel siding	26.0000	480.0000	0.1000	0.0050	0.0002	
14	271	2x4@24" oc + R11 Batt	0.0280	7.1100	0.2000	0.2917	10.4179	
15	94	BUILT-UP ROOFING, 3/8 IN	0.0930	70.0000	0.3500	0.0313	0.3366	
16	21	4 in. Insulation	0.0250	5.7000	0.2000	0.3330	13.3500	
17	9	Air space resistance	0.0000	0.0000	0.0000	0.0000	0.9100	✓

The following columns may be edited by the user:

Construct Grid - Constructs may be added or removed by following the general procedures for adding and deleting rows from grids.

Simple Construct - Beginning version 1.22, the 'Simple Construct' option is available to the user when the user decides to enter gross properties of the construct rather than through individual layers. When the 'Simple Construct' option is checked for a particular construct, all layers, if any, for that construct are ignored. The user will now be able to enter the gross properties (Conductance, Heat capacity, Density and Thickness) for that construct in the construct grid.

No-Mass Construct - Beginning version 1.22, the 'No-Mass Construct' option is available to the user when the user decides that the wall is too thin or too light for the thermal mass have any impact on compliance calculations. When the 'No-Mass Construct' option is checked for a particular construct, only the U-Value is considered in the calculation. This option should be used if most materials that make up the construct are defined only using R-Value for the materials or if the gross thickness of the construct is less than an inch (0.084 ft). In general, use this option if mass-related properties are not available for the gross construct or the materials that constitute the layers of the construct.

Outside Absorptance - Specifies the fraction of thermal radiation absorbed by the outside layer. This can take values between 0.0 and 1.0.

Inside Emittance - Specifies the fraction of thermal radiation emitted by the inside layer. This can take values between 0.0 and 1.0.

User may also modify the entries under the 'Construct' column of the grid. Type a names that makes the most sense for the construct.

All other fields in this grid are not editable, except when the construct is declared '**Simple**'. They are automatically recalculated and updated based on changes to layers and/or materials when '**Simple Construct**' is unchecked.

Conductance - Specifies the overall thermal conductivity of the construct. Conductance may be entered if you check the Simple Construct option. Otherwise the field is non editable and is automatically calculated from the layer data.

Heat Capacity - Specifies the over all heat capacity of the construct. Heat Capacity may be entered if you check the Simple Construct option and uncheck the No-Mass construct option. Otherwise the field is not editable and is automatically calculated from the layer data.

Density - Specifies the overall density of the construct. Density may be entered if you check the Simple Construct option and uncheck the No-Mass construct option. Otherwise the field is not editable and is automatically calculated from the layer data.

Thickness - Specifies the overall thickness of the construct. Thickness may be entered if you check the Simple Construct option and uncheck the No-Mass construct option. Otherwise the field is not editable and is automatically calculated from the layer data.

R-Value - Specifies the overall R-Value of the construct. This field is not editable and is calculated based on other inputs

NOTE: Constructs that may be problematic are flagged as shown for Construct No. 1047 above. If this construct is used in a project envelope component, running '**Calculate**'>'**Error Check**' will provide error information possible resolution.

Layer Grid

Layers may be added or removed by following the general procedures for adding and deleting rows from grids.

User may change the material of any layer by clicking on the cell button that appears when any cell under the material column in the layer grid is selected.

When a layer is added, the thickness defaults to the thickness of the material selected for the layer. User may override the thickness by typing a different value.

User may change the framing factor by typing the desired value. Framing factor is the area-fraction of the framing in a layer. That is, it is the ratio of the area of framing to the total area of the wall. It should be entered for those layers mounted on framing. For example, insulation that is mounted on framing.

User may reorder the layers in the layer grid by holding left mouse button down on the label cell (the cells in the left most column) of the layer to be moved, and dragging it up or down. A drag icon appears during the dragging process.

See the section on [Working with Grids](#) to add, delete, or modify data.

Materials Tab

The Project Materials Library is very similar to the Master Materials Library except that it is associated with a specific Project that is currently open or created. This library stays with the Project and changes made here are applicable only to the current project.

The user may make any modification as required.

Constructs		Materials						
Mat No	Desc	Conductivity [Btu/h.ft.F]	Density [lb/cf]	SpecificHeat [Btu/lb.F]	Thick [ft]	R-Value [h.ft.F/Btu]	Use R-Value	
1	264 ALUMINUM, 1/16 IN.	26.0000	480.0000	0.1000	0.0050	0.0002	☐	
2	214 POLYSTYRENE, EXP., 1-1/4 IN.	0.0200	1.8000	0.2900	0.1942	5.2100	☐	
3	187 GYP OR PLAS BOARD, 1/2 IN.	0.0920	50.0000	0.2000	0.0417	0.4533	☐	
4	178 CARPET W/RUBBER PAD	0.0000	0.0000	0.0000	0.0000	1.2300	☒	
5	265 Sol. 1 ft	0.5000	100.0000	0.2000	1.0000	2.0000	☐	
6	49 6 in. Heavyweight concrete	1.0000	140.0000	0.2000	0.5000	0.5000	☐	
7	267 0.75" stucco	0.4000	16.0000	0.2000	0.0625	0.1563	☐	
8	215 POLYSTYRENE, EXP., 2 IN.	0.0200	1.8000	0.2900	0.1667	8.3350	☐	
9	105 CONC BLK HW, 8 IN, HOLLOW	0.6060	69.0000	0.2000	0.6867	1.1002	☐	
10	256 WOOD, SOFT, 1-1/2 IN.	0.0660	32.0000	0.3300	0.1250	1.8539	☐	
11	86 BRICK, COMMON, 4 IN.	0.4160	120.0000	0.2000	0.3333	0.8012	☐	
12	23 6 in. Insulation	0.0250	5.7000	0.2000	0.5000	20.0000	☐	
13	4 Steel siding	26.0000	480.0000	0.1000	0.0050	0.0002	☐	
14	271 2x4@24" oc w R11 Batt	0.0200	7.1100	0.2000	0.2917	10.4179	☐	
15	54 BUILT-UP ROOFING, 3/8 IN.	0.0930	70.0000	0.3500	0.0313	0.3366	☐	
16	21 4 in. Insulation	0.0250	5.7000	0.2000	0.3330	13.3200	☐	
17	9 Air space resistance	0.0000	0.0000	0.0000	0.0000	0.9100	☒	

Materials may be added, removed, and changed by following the general procedures for working with grids.

All fields in the grid except 'Mat No.' can be changed by the user.

Conductivity - Specifies the thermal conductivity of the material

Density - Specifies the material density

Specific Heat - Specifies the specific heat of the material

Thick - Specifies the thickness of the material in feet

R-Value - Specifies the thermal resistance of the material

Use R-Val - Two possible specification for a material are possible:

- 1) Specify valid thickness and all thermal properties. In this case, valid conductivity, density and specific heat must be specified and the 'Use RVal' check box must be unchecked. This option is used for materials that are likely to have mass effects or when these properties are accurately known. The program will use the thermal properties and ignore the R-Value
- 2) For light and thin materials, or for those materials for which the thermal properties are not available, the second option is to specify only the R-Value. In this

case the 'Use R-Value' option must be checked. The program will ignore the thermal properties, and use only the R-Value and treat the material as having no mass.

IMPORTANT CAUTIONARY NOTE: It is critical to note that any change to the material properties, will affect the properties of all constructs that use this material as a layer. In general changes to Project Library of materials, constructs or fenestration, will propagate throughout the entire project which use this library component.

See the section on [Working with Grids](#) to add, delete, and modify data.



5.2 Project Fenestrations Library Form

The Project Fenestration library allows the user to add Fenestrations to the Project, which can then be selected by a project window or skylight as the fenestration type.

	No	Name	Description	No of Panes	Glass Cond [Btu/h.SF.F]	SHGC	VLT	Frame Type
1	1	ASHULSqClrAlFrm	ASH90.1 Unlabeled Sq Clear All frame types	1	1.250	0.820	0.760	Metal Frame (Generic)
2	12	ASHULSkyL0.125GlsAlWvTbrk	ASH90.1 Unlabeled Skylight .125 in glass Al frm	1	1.980	0.820	0.760	Metal Frame (Generic)
3	277	AplBwnd277	T11B1C90VF:N:0:	1	2.000	0.200	1.000	Metal Frame (Generic)
4	278	AplBwnd278	T11B1C90VF:A:0:	1	0.670	0.390	1.000	Metal Frame (Generic)
5	305	AplBwnd305	T11B1C61SK:NC:	1	0.690	0.490	1.000	Metal Frame (Generic)
6	306	AplBwnd306	Window 306	1	1.250	0.200	1.000	Metal Frame (Generic)

No. - Specifies the ID number associated with the fenestration

Name - Specifies a name for the fenestration. User can type any convenient, short description

Description - A detailed description that can be up to 50 characters

No. of Panes - Specifies the number of glass panes for the user defined fenestration

Glass Conductance - Specifies the conductance of glass panes for the user defined fenestration

SHGC - Specifies the solar heat gain coefficient of the glass panes for the user defined fenestration

VLT - Specifies the Visible Light Transmittance

Frame Type - Describes the type of frame used

See the section [Working with Grids](#) for instructions to add, delete, or modify data.



6. Reference Manuals

6.1 Florida Building Code

The code manuals are copyrighted by a separate entity and must be ordered separately. For more information on how to obtain code manuals and CDs, contact:

International Code Council:

Address:

500 New Jersey Avenue, NW, 6th Floor

Washington, DC 20001-2070

USA

Telephone: 1-888-422-7233

Website: <http://www.iccsafe.org>

Also, this [website](#) has links to online documentation and downloads (including forms) for the Florida Building Code.

7. Technical Support

7.1 System Requirements

To run *EnergyGauge Summit*, your system must meet the following minimum requirements:

- 1 gigahertz (GHz) or faster 32-bit (x86) or 64-bit (x64) processor*
- 2 GB RAM
- 1 GB available hard disk space
- DirectX 9 graphics device with WDDM 1.0 or higher driver
- .NET 3.5 Framework
- Internet access for activation

- Windows 7 or later

*Based on your specific OS version and other software installed on your machine, it is possible that some conflicts may arise. These have to be resolved on a case-by-case basis. Please report any problems you encounter promptly to the EnergyGauge Technical Support Office.

7.2 Activation and Support

Activation:

EnergyGauge Summit has to be activated either through the Internet or by phone after installation to be able to use it.

Technical Support:

Software Support - *EnergyGauge Summit* is only the first of a suite of software products for commercial building energy analysis to be developed by the Florida Solar Energy Center. Follow-up versions will offer continuous improvement of features and customized results to meet user needs.

Support Website - Please visit our [Technical Support Website](#) to check for updates, FAQs, downloads, and knowledge base articles.

Training Support - Training in a variety of renewable energy and energy-efficiency subject areas is available through the Florida Solar Energy Center at their [Education & Training](#) page.

E-mail Support - You may contact the EnergyGauge Support Office by [e-mail](#). Please include your phone number in case we need to call you back, and **be sure to include your full software version number in the subject line of all e-mail questions (see main 'Help'>'About' screen in your software for your full version number).**

Phone Support - If all else fails or you have what you consider to be an emergency, the EnergyGauge Support Office may be reached by phone at (321)-323-7255.

The mailing address for the EnergyGauge Support Office is:

1679 Clearlake Road,
Cocoa, FL 32922-5703